

White Label Farm App

Build Specification

Version 1.1 · 3 August 2026

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1 Summary

1.1 At a glance

Product	White label farm management application, built on Map My Crop's satellite analytics platform
Markets	GCC and Jordan
Platforms	iOS 15+, Android 8.0+. Mobile only
Languages	English, Arabic, Hindi, Bengali, Pashto — all at launch. Arabic and Pashto right-to-left
Roles	Farm Owner, Farm Supervisor, Farm Worker — three templates on a capability matrix
Plans	Crops: Basic → Pro → Advanced. Trees: Basic → Pro. Combined plans where a client holds both
Payment	Self-signup, in-app purchase through the App Store and Google Play
Hosting	Data residency to be agreed — see open items
Offline	Read cached data anywhere; capture tasks, photos, observations and applied inputs offline
Build	One release, no phasing. Two months from sign-off

1.2 What already exists and what must be built

The crop feature list and the tree monitoring catalogue describe capabilities that are built and in use. This specification takes them as given.

Area	Status	Notes
Satellite measures, health scores, per-tree analytics	Exists	Consume as-is
Weather, forecast, historical data, ET/ET0	Exists	
Disease directory, forecasting, photo detection	Exists	
Irrigation scheduling	Exists	Advanced crop plan; Tree Pro
Fertiliser insights	Exists	Pro and above
Farm diary, scouting, dashboards, reports	Exists	Rebrand only
VRA maps, advanced GIS layers, compare maps	Exists	
Expert opinion, Agro Doctor, support tickets	Exists	Route to Wafra first line
Branding and app identity	New — small	

Area	Status	Notes
In-app contact buttons (email, WhatsApp)	New — small	
Five languages with full RTL	New — large	Current product is effectively English-only
Farm sharing and role-based access	New — large	Current product is single-user
Farm → block → field → plot hierarchy, multi-crop	New — assuming a core data model change	Current product is one crop per plot
Crop cycle as a first-class time-bounded entity	New — assuming a core data model change	
Task management	New — large	No equivalent in the current feature lists
Advised-vs-applied recording and reporting	New	
Offline capture and sync	New — large	
Entitlements, trial, in-app purchase	New	Including combined crop + tree plans
Immutable advisory log	New	Cannot be retrofitted
Registered plant protection product register, per country	New	The disease directories describe diseases, not products, rates or pre-harvest intervals

2 General rules

2.1 Requirement identifiers

Every requirement carries an ID in the form **WF-*nnn***, numbered in the order they appear. Use these IDs in estimates, in commit messages, in test cases and in acceptance sign-off. The full register is in the last annex.

Chapter	Requirements
2 General rules	WF-001 – WF-017
3 Information architecture and navigation	WF-030 – WF-034
4 Onboarding and access	WF-100 – WF-170
5 Screen specifications	WF-200 – WF-345
6 Advisory layer	WF-600 – WF-624
7 Notifications	WF-650 – WF-657
8 Roles and permissions	WF-670 – WF-678
9 Plans and entitlements	WF-700 – WF-715
10 Internationalisation (i18n)	WF-750 – WF-768
11 Offline behaviour	WF-780 – WF-792
12 Deployment and release	WF-800 – WF-816
13 Acceptance	WF-850 – WF-852
Annex A Hosting, data and security	WF-900 – WF-906

The first digits of an ID identify the chapter it belongs to. Wireframes are layout and hierarchy instructions, not visual design.

2.2 Terminology

Term	Meaning
Farm	The top-level land holding. Has a name, a location and a type.
Block	An optional subdivision of a farm, usually by geography or irrigation system.
Field	A managed unit inside a block.
Plot	The smallest mapped polygon. Carries a boundary. Analytics are computed per plot.
Crop cycle	A time-bounded planting of one crop in one plot, from sowing to harvest. A first-class record with its own history — not an attribute of the plot.
Measure	A satellite-derived vegetation or health value (NDVI, NDRE, EVI, MSAVI and others). Called <i>measure</i> everywhere in the interface; the technical name appears only in the legend and in reports.

Term	Meaning
Recommendation	A piece of advice produced by the platform and shown to the user.
Task	A unit of work assigned to a person, with a due date and a completion record.
Observation	A field-captured note, usually with a photo and a GPS position.
Entitlement	The server-held record of what an account is allowed to see and do.
Plan	The subscription a client holds.

2.3 Global rules

These apply to every screen and are part of acceptance.

WF-001 — Mobile application for **iOS 15+** and **Android 8.0 (API 26)+**. No browser application in this release.

WF-002 — Usable on a 5-inch screen at 360 × 640 dp. Every screen is tested at that size before it is considered done.

WF-003 — Install size below 80 MB.

WF-004 — Minimum touch target 48 × 48 dp, with at least 8 dp between adjacent targets.

WF-005 — The primary action on every screen sits in the bottom third of the display. Destructive or rarely-used actions go to the top.

WF-006 — Body text no smaller than 16 sp. Numbers the user acts on (water volume, dose, tree count) no smaller than 20 sp.

WF-007 — Respect the operating-system font-size setting up to 200% without clipping text or losing buttons off-screen.

WF-008 — Status is never communicated by colour alone. Every colour-coded state also carries an icon and a text label.

WF-009 — Health and status use this four-state scale everywhere, with the same icon, word and colour every time:

State	Icon	Label (EN)	Meaning
Good	filled circle	Good	No action needed
Watch	half circle	Watch	Monitor, no action yet
Act	triangle	Action needed	Act within days
Urgent	filled triangle	Urgent	Act today
No data	dashed circle	No data	Cloud cover, no imagery, or outside subscription

WF-010 — No screen presents more than one primary action.

WF-011 — Every screen that can legitimately have nothing to show — no farms yet, no plots in a block, no tasks today, no advice outstanding, no imagery for the selected date — has a designed empty state: an icon, one line of plain language saying why it is empty, and a button that resolves it where one exists ("Add your first farm"). An empty list rendered as a blank area is a defect.

WF-012 — Every screen has an explicit loading state and an explicit error state with a plain-language message and a *Try again* button. Log the error code, show the sentence.

WF-013 — Numbers are always shown with their unit, rounded to actionable precision: "412 m³" not "411.8734 m³".

WF-014 — Icons carry a text label beneath or beside them.

WF-015 — Never block the user with a full-screen "no connection" message when cached data exists. Show the cached data with its age and let the user continue.

WF-016 — Every data-entry action that can be performed offline shows immediate local confirmation.

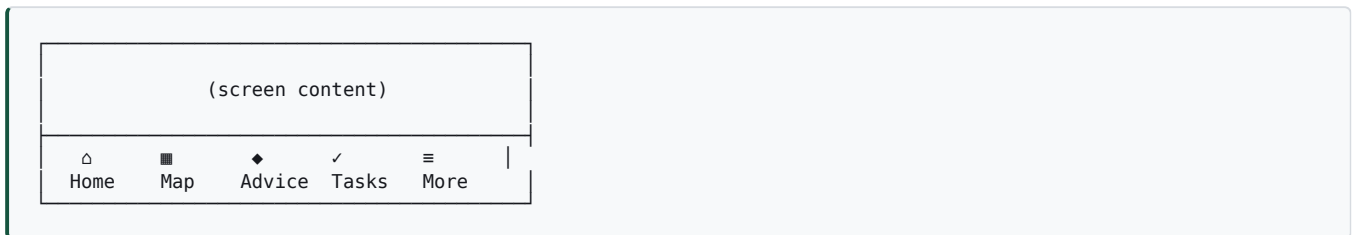
WF-017 — The application carries Wafra branding only — logo, colours, app icon, splash, store listing and generated reports. No MMC logo, copyright line, support address or brand name appears anywhere in the app, in reports, in email, in push notifications or in the store listings. Colour is applied through named design tokens, not hard-coded values.

3 Information architecture and navigation

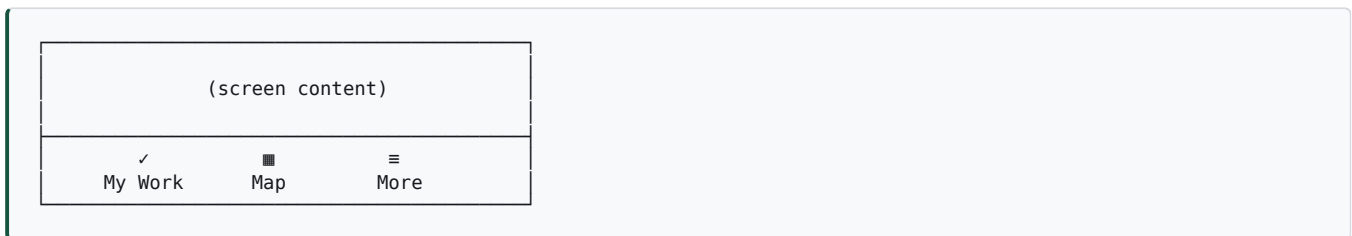
3.1 Global navigation

Navigation is a bottom tab bar, always visible, never hidden on scroll. Tabs are role-adaptive.

WF-030 — Owner and Supervisor see five tabs:



WF-031 — Worker sees three tabs, and lands on My Work at launch:

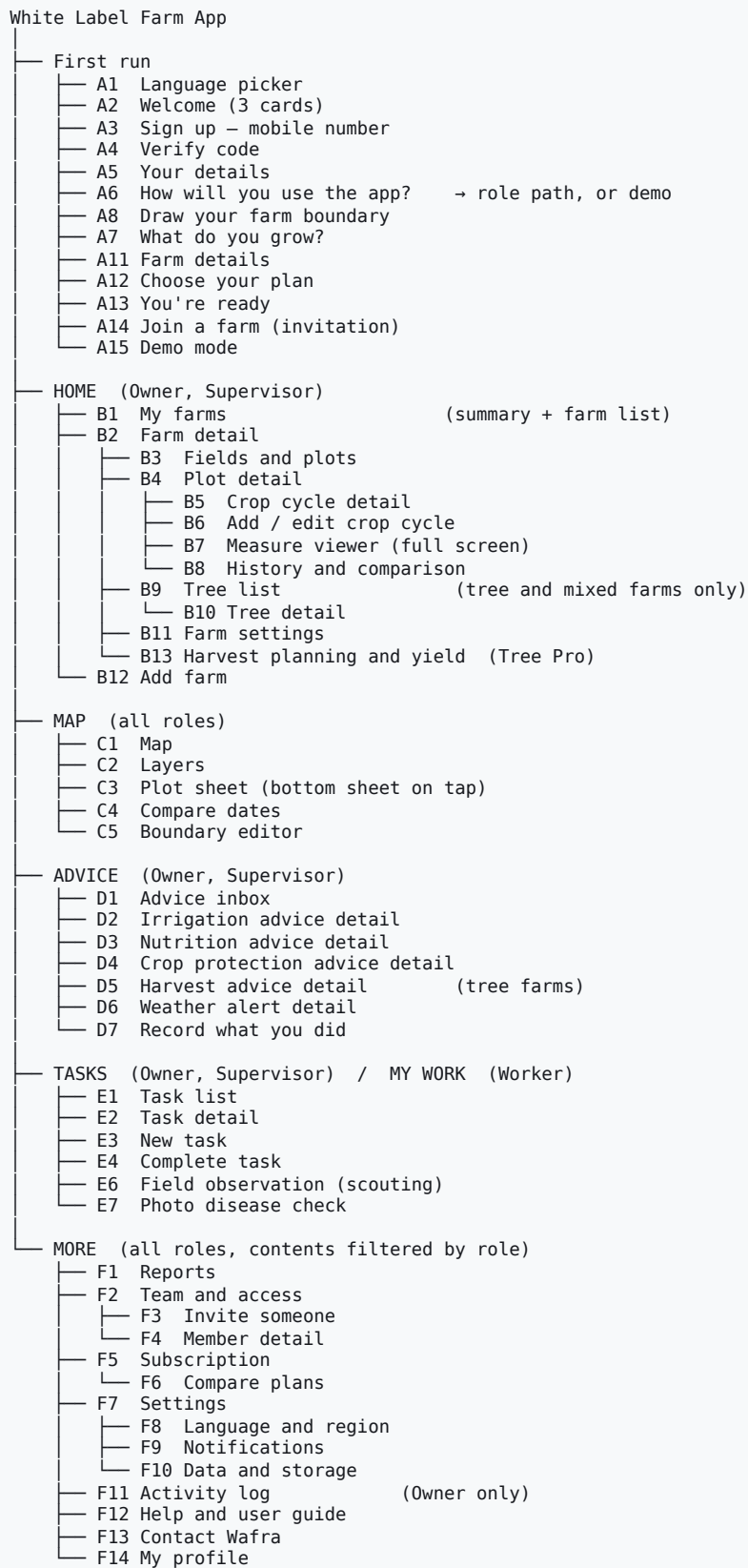


WF-032 — A badge on the Advice tab shows the count of unread recommendations at **Action needed** or **Urgent** severity only. Routine information is never badged.

WF-033 — A badge on the Tasks / My Work tab shows the count of tasks assigned to this user and due today or overdue.

WF-034 — Tab bar labels are always visible.

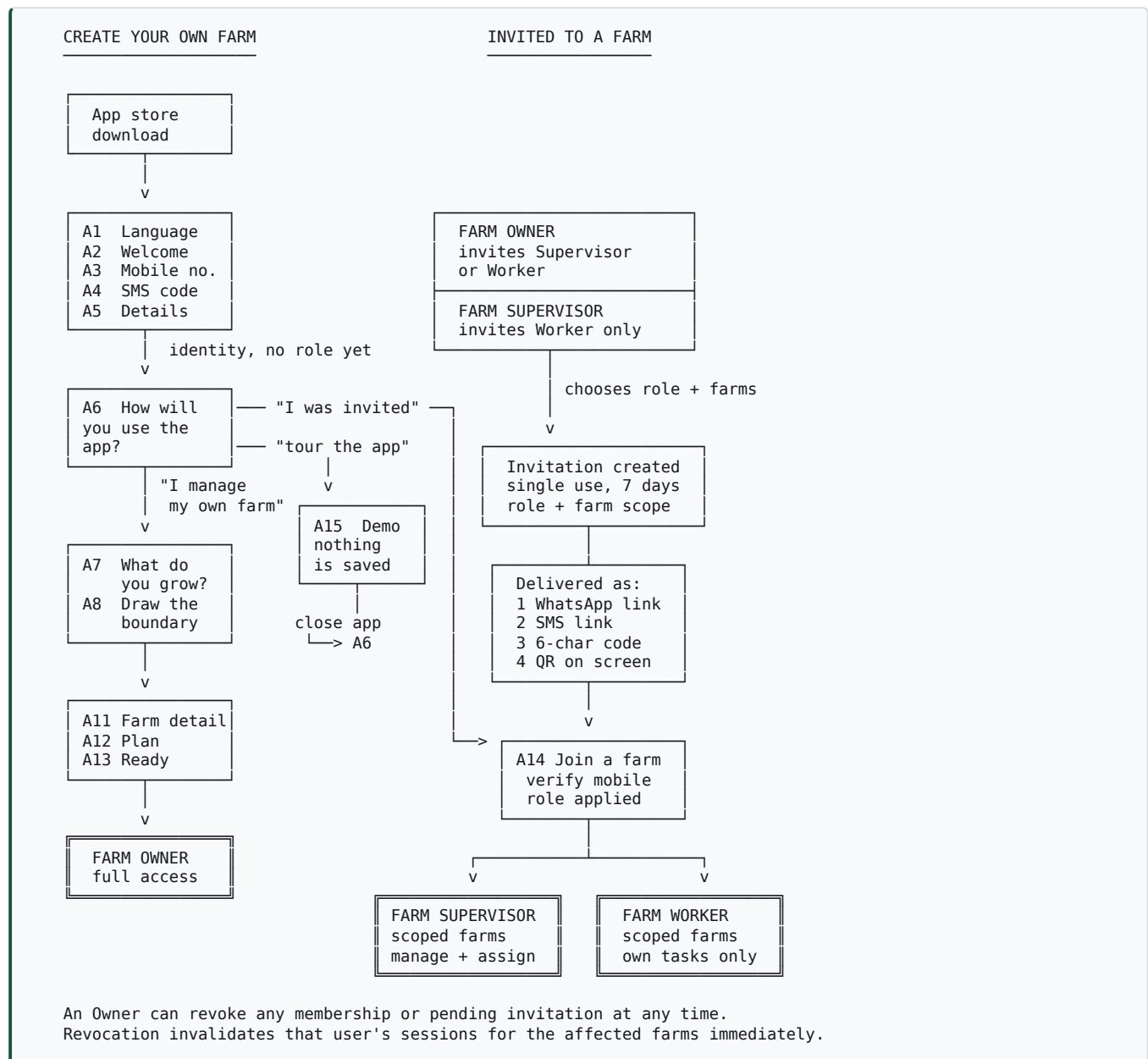
3.2 App Map



4 Onboarding and access

4.1 Registration and access model

Everyone registers the same way: mobile number, SMS code, name. Registration creates an identity, not a role. What happens next is chosen on one screen (A6), and that choice determines the role.



4.2 How roles are assigned

WF-100 — Registration creates an account with **no role**. A role is attached to a user only through a farm: either by creating one, or by accepting an invitation to one.

WF-101 — A user who creates a farm becomes a **Farm Owner** of it.

WF-102 — A user who accepts an invitation receives the role carried in that invitation — **Farm Supervisor** or **Farm Worker**. The role is set by the inviter and is never chosen by the invitee.

WF-103 — The same person may do both. Registering, creating a farm, and later accepting a Supervisor invitation to someone else's farm is a normal case: roles are held per farm, not per account.

WF-104 — A Worker can never create, delete or edit a farm, a plot or a boundary, on any farm, however they arrived.

WF-105 — An invitation is single use, expires after 7 days, and is bound to the role and the farm scope chosen at creation.

WF-106 — Four delivery methods, all producing the same single-use token: WhatsApp deep link (primary), SMS deep link (fallback), a 6-character code the invitee types manually, and a QR code displayed on the inviter's screen.

WF-107 — An Owner sees all pending invitations for their farms with expiry and a one-tap revoke. A Supervisor may invite Workers only, and only to farms they already have access to.

WF-108 — Where a Worker has no smartphone of their own, a Supervisor can complete the invitation on the Worker's device by entering the 6-character code.

4.3 A1 — Language picker

Shown on very first launch only.



[LOGO]

Choose your language
(repeated in Arabic below it)

[native name]	English	✓
[native name]	Arabic	
[native name]	Hindi	
[native name]	Bengali	
[native name]	Pashto	

Continue

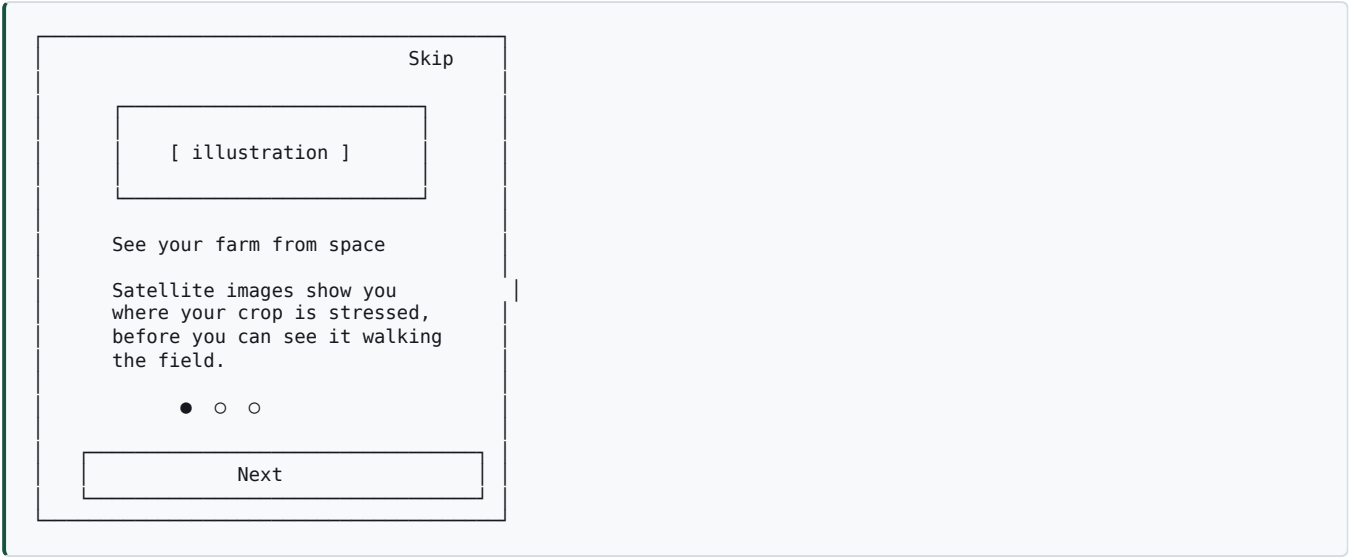
WF-109 — Each language is shown in its own script, with the English name beside it.

WF-110 — Pre-selected to the device's OS language if it is one of the five; otherwise English.

WF-111 — Selecting a right-to-left language mirrors the layout immediately, on this screen, before Continue is pressed.

WF-112 — Never shown again after first run. Changed afterwards in Settings (F8) and on the login screen.

4.4 A2 — Welcome



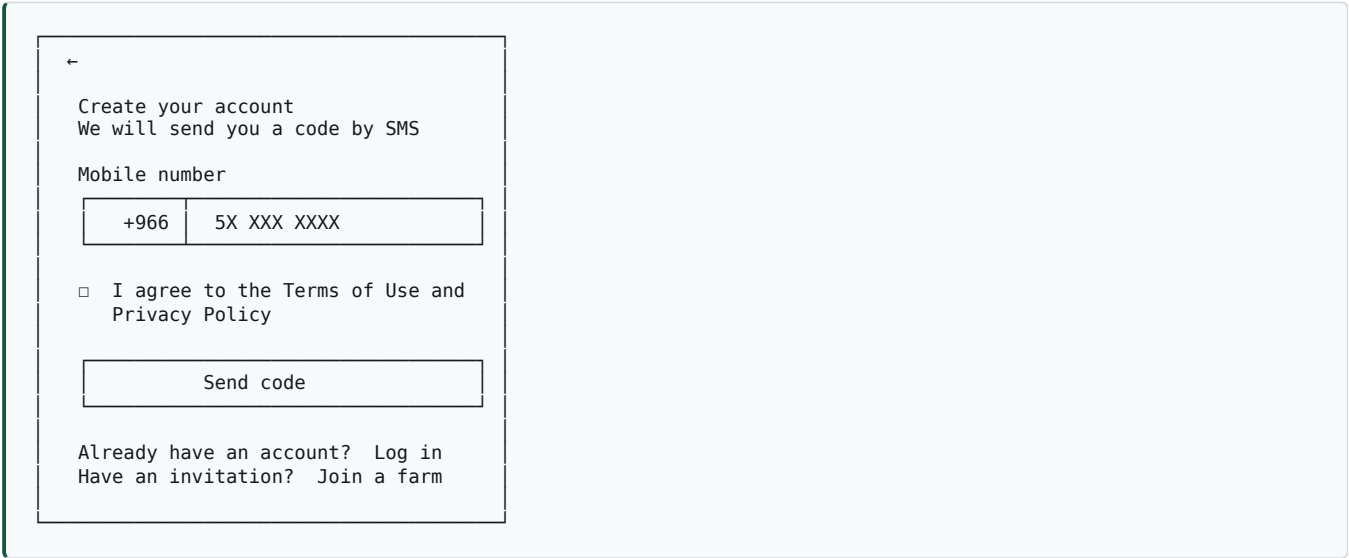
Three swipeable cards with a visible *Next* button and a *Skip* control on every card.

Card	Headline	Body
1	See your farm from space	Satellite images show you where your crop is stressed, before you can see it walking the field.
2	Know what to do	Get irrigation, nutrition and pest advice for your crops, in your language.
3	Work as a team	Send jobs to your workers and see them done, with photos.

WF-113 — Three cards maximum, skippable from any card.

4.5 A3–A4 — Sign up

WF-114 — Registration is by mobile number with SMS one-time code. Mobile number is the primary account identifier. Email is optional.



WF-115 — Country selector defaults from the SIM's country, then device locale, then Saudi Arabia. The GCC states and Jordan appear at the top of the list.

WF-116 — The number field accepts spaces and dashes and normalises them. Leading zeros after the country code are stripped: 0501234567 with +966 selected must succeed.

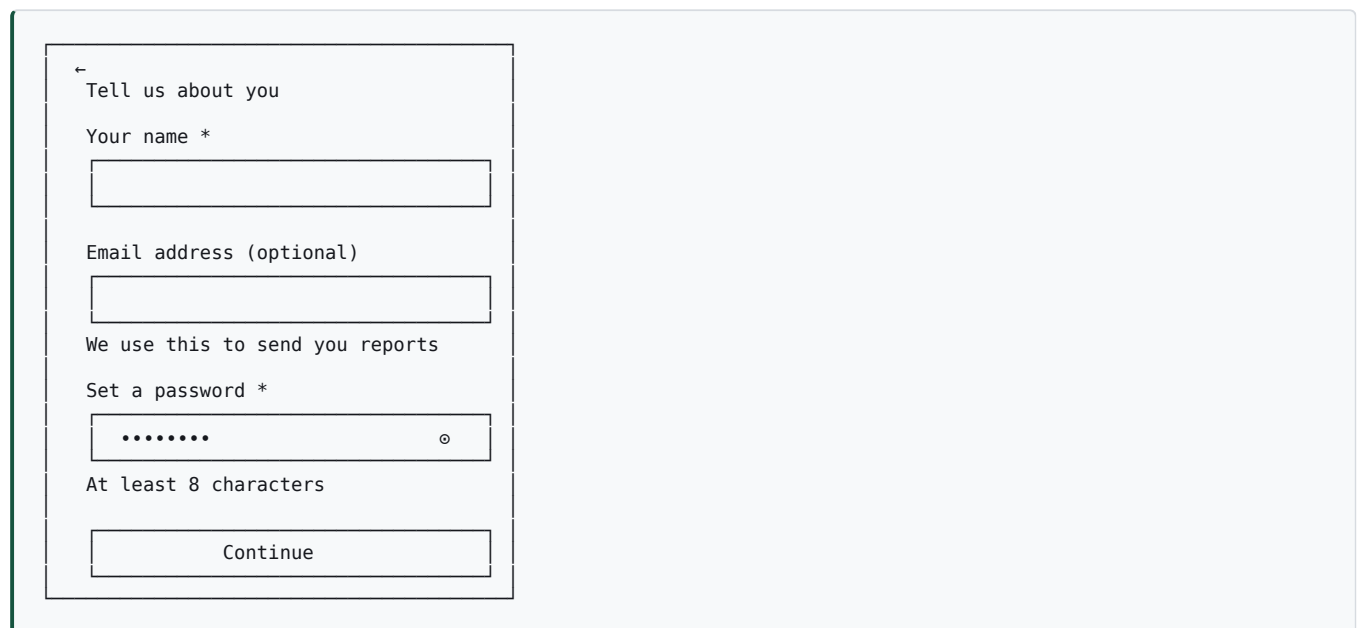
WF-117 — Terms and Privacy links open in-app, in the user's selected language. The checkbox is unticked by default and Send code is disabled until it is ticked.

WF-118 — Six-digit code, valid for 10 minutes. Auto-read from SMS on Android (SMS Retriever API) and iOS autofill. Auto-submits at six digits.

WF-119 — Resend available after 45 seconds. Maximum 5 code requests per number per hour, enforced server-side.

WF-120 — After 5 incorrect code entries the account is locked for 15 minutes and the user is shown the Wafra contact options (F13).

4.6 A5 — Your details

A screenshot of a mobile app registration screen titled "Tell us about you". The form is contained within a white box with a thin border. It includes a back arrow at the top left. The fields are: "Your name *" (mandatory), "Email address (optional)", "Set a password *" (mandatory), and a "Continue" button at the bottom. The password field has a show/hide toggle icon on the right. Below the password field, the text "At least 8 characters" is displayed. The background of the app screen is a light blue-grey color.

WF-121 — Name is mandatory; email optional. Password minimum 8 characters, no composition rules, checked against a list of the 10,000 most common passwords.

WF-122 — A show/hide control on the password field.

WF-123 — Password is optional for Worker accounts created by invitation; those may sign in by SMS code alone. Owners and Supervisors must set one.

4.7 A6 — How will you use the app?

← How will you use the app?

△
I manage my own farm
Set up your farms and invite
your team

✓
I was invited to a farm
Enter your invitation code or
scan the QR code

■
I just want to tour the app
Look around with example data.
Nothing is saved.

WF-124 — This screen routes, it does not assign privileges. "I manage my own farm" continues to A7 and the user becomes Owner of the farm they create. "I was invited" continues to A14 and the role comes from the invitation.

WF-125 — A user who arrives through an invitation deep link skips A6 entirely.

WF-126 — *I just want to tour the app* opens a **demo session**: a read-only example farm with sample imagery, advice and tasks, served from a fixture, in the user's chosen language. It is bound to the app process — closing the app discards it, and reopening returns the user to A6.

WF-127 — A demo session writes nothing to the server, creates no account, consumes no entitlement and cannot be shared or invited into. A persistent banner reads *Demo — nothing you do here is saved*, with an *Exit demo* control that returns to A6.

WF-128 — Every action in a demo session that would normally write data completes locally and visibly, then resets with the session. Actions are never disabled.

WF-129 — The demo fixture is delivered by the backend configuration endpoint so its content can be updated without an app release.

4.8 A7 — What do you grow?

←
What do you grow?
You can change this later

*
Field crops
Wheat, alfalfa, potato,
vegetables, fodder

♣
Trees and orchards
Date palm, olive, citrus,
mango, grapes

* ♣
Both
You will be offered a combined
plan

Asked once, immediately before the first farm is created.

WF-130 — Three options — Field crops, Trees and orchards, Both — setting the default `farm_type` for the first farm and determining which plan family is offered at A12. It does not restrict the account; a crop client adding a tree farm later triggers a tree plan offer.

4.9 A8 — Add your first farm

←
Add your farm

✎ Draw it on the map
Trace the boundary on satellite
imagery

WF-131 — This screen is reachable only by Owners and Supervisors with the `plot.create` capability. It never appears for a Worker.

4.9.1 A8 — DRAW IT ON THE MAP



- WF-132** — Opens centred on the device's GPS position at approximately 1:5,000. If location permission is refused, opens on a country-level search box.
- WF-133** — Satellite basemap by default.
- WF-134** — Live area readout in dunum with hectares in brackets. See §10.4.
- WF-135** — *Undo* removes the last placed vertex. *Clear all* is available behind a confirmation.
- WF-136** — Vertices are draggable with a touch target of at least 48 dp, offset above the fingertip so the dragged point is not hidden by the finger.
- WF-137** — Self-intersecting polygons are rejected with a plain-language message and the offending segment highlighted.
- WF-138** — A polygon smaller than 0.1 ha or larger than 10,000 ha triggers a confirmation, not a rejection.

4.10 A11 — Farm details

← Farm details

Farm name *

Al Kharj North

What is on this land? *

() Field crops

(●) Trees and orchards

() Both

Irrigation system *

Drip ▼

Soil type

Sandy loam ▼

Not sure? We will estimate it

Save farm

WF-139 — Only farm name and farm type are mandatory. Every other field is optional and can be completed later.

WF-140 — Irrigation system options: Drip, Centre pivot, Sprinkler, Bubbler, Flood/furrow, Other, Not sure. "Not sure" is a valid answer and blocks nothing.

WF-141 — Soil type may be left blank; the backend fills it from the soil layer where available and the app shows it as an estimate the user can override.

4.11 A12 — Choose your plan

WF-142 — The plan chooser shows only the plan family matching the farm type just created.



WF-143 — Plan order is always Basic → Pro → Advanced for crops and Basic → Pro for trees. Pro is presented before Advanced.

WF-144 — Prices are fetched from the server, never hard-coded, and displayed in the currency of the user's country. USD is not shown to the farmer. See §10.4.

WF-145 — *Compare all features* opens the full comparison table (F6).

4.11.1 WHEN A CLIENT HAS BOTH CROPS AND TREES

Crops and trees are analysed differently and are licensed differently — crop capability follows the farm, tree capability follows the tree count. A client holding both must not be asked to buy and manage two subscriptions.

WF-146 — Where the account has at least one crop farm and at least one tree farm, the plan chooser offers **combined plans** and does not offer the two families separately:

Combined plan	Grants
Complete Basic	Crop Basic on every crop farm, Tree Basic on every tree farm
Complete Pro	Crop Pro on every crop farm, Tree Pro on every tree farm
Complete Advanced	Crop Advanced on every crop farm, Tree Pro on every tree farm

WF-147 — A combined plan is a **single purchasable product** with a single price and a single renewal date. The app never presents it as two line items.

WF-148 — A mixed farm — crops and trees in different plots of the same farm — requires a combined plan. The app states this at A11 when *Both* is selected as the farm type.

WF-149 — Where a client on a crop-only plan adds their first tree farm, the app offers an upgrade to the matching combined plan rather than a second subscription. The reverse applies when a tree-only client adds a crop farm. Until they upgrade, the new farm is visible but its analytics are locked, with the upgrade path shown in place of the data.

WF-150 — Entitlement is resolved server-side as a single set of feature keys. The app asks what the user may do and never derives capability from the plan name or the farm type (§9.5).

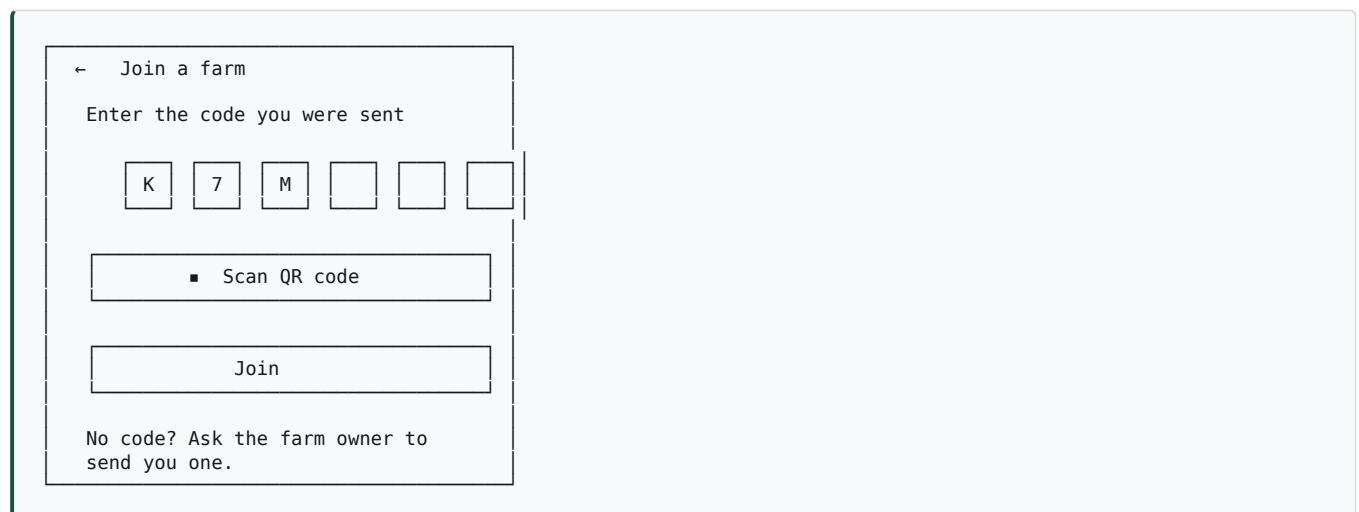
WF-151 — Downgrading from a combined plan to a single family is only possible once the farms of the other type have been archived. The app explains which farms block the change and offers to archive them.

4.12 A13 — You're ready



WF-152 — A confirmation screen stating when the first satellite imagery will be available, and a single button to Home.

4.13 A14 — Join a farm



WF-153 — Opening an invitation deep link, with or without the app installed, leads to: install if needed → verify mobile number → land inside the farm they were invited to, with their role already applied.

WF-154 — An invited user never sees A6, A7–A11 or A12. They join an existing subscription.

WF-155 — Invitations expire after 7 days and are single use. See §4.1.

WF-156 — A used, expired or revoked invitation shows a clear message and a *Contact the farm owner* action. It never fails silently and never grants partial access.

WF-157 — On successful join, a Worker lands directly on My Work; a Supervisor lands on Home filtered to the farms they were granted.

4.14 Login and recovery

The mockup shows a login screen with the following elements:

- Language selector: English (with a dropdown arrow)
- [LOGO]
- Mobile number label
- Mobile number input field: +966 | 5X XXX XXXX
- Send me a code button
- or separator
- Password label
- Password input field: (with an eye icon for toggling visibility)
- Log in button
- Links at the bottom:
 - Forgot your password?
 - New here? Create an account
 - Have an invitation? Join a farm

WF-158 — Login accepts mobile number + password, or mobile number + SMS code. Both are offered on the same screen; SMS code is the more prominent option.

WF-159 — A language control is present on the login screen.

WF-160 — Sessions persist for 90 days of activity on a personal device. Biometric unlock is offered where the device supports it.

WF-161 — A shared device mode toggle in Settings shortens the session to 12 hours and requires re-authentication on app launch.

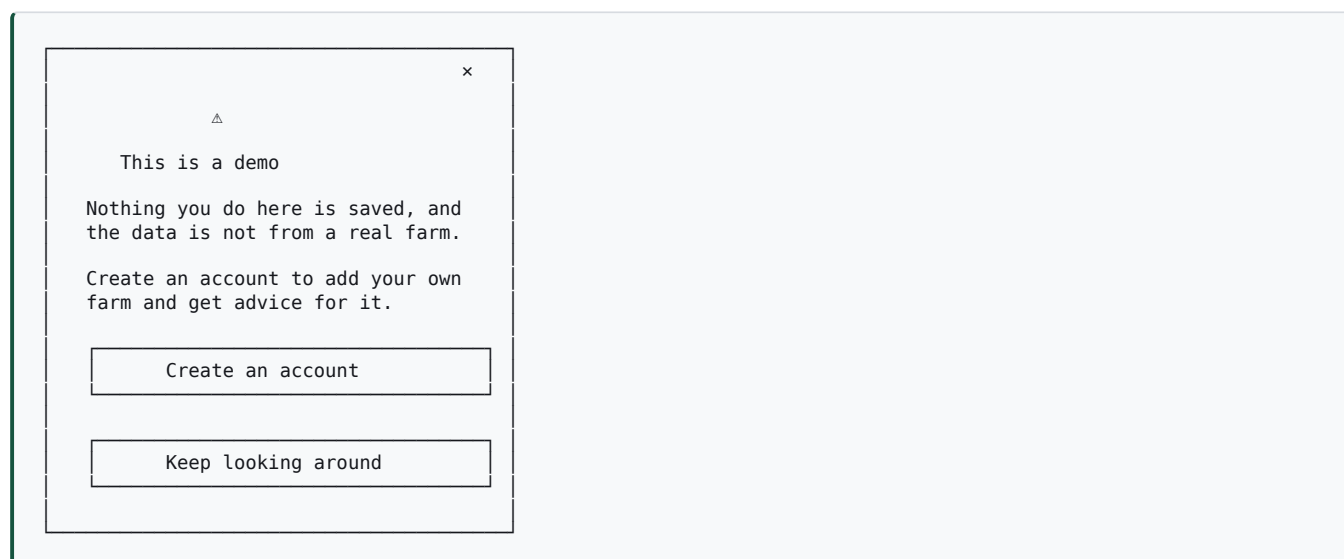
WF-162 — *Log out* is always available and clears the local cache including cached imagery and queued photos, after warning the user if anything is unsynced.

4.15 A15 — Demo mode

Reached from A6 by *I just want to tour the app*. A demo session lets someone see what the product does before committing a mobile number, and gives a sales conversation something to point at.



Tapping *Exit* or attempting anything that would need an account:



WF-163 — A demo session opens a fixture: two example farms — one crop, one tree — with plots, imagery for the last 60 days, advice, tasks and a team, in the user's chosen language.

WF-164 — A demo session writes nothing to the server, creates no account, consumes no entitlement, and cannot be invited into or shared.

WF-165 — The banner at the top of every screen reads *Demo — nothing here is saved* and carries an *Exit* control. It is not dismissible.

WF-166 — Every action that would normally write data completes locally and visibly, then resets with the session. Controls are never disabled.

WF-167 — Any action that genuinely requires an account — subscribing, inviting a team member, adding a real farm — opens the conversion sheet above rather than failing.

WF-168 — The session is bound to the app process. Closing the app discards it, and reopening returns the user to A6. There is no way to return to a previous demo session.

WF-169 — All plan-gated features are shown **unlocked** in the demo, so a prospect sees what the product does rather than what they have not bought. Every advisory screen still carries its disclaimer (§6.7).

WF-170 — The demo fixture is delivered by the backend configuration endpoint, so its content can be updated without an app release.

5 Screen specifications

5.1 B1 — Home / My farms

The first screen after login for Owners and Supervisors.



WF-200 — The summary card at the top counts farms by worst-plot status, not by average. A farm with nine healthy plots and one urgent plot is an urgent farm.

WF-201 — *See what to do* navigates to the Advice inbox (D1) pre-filtered to Action needed and Urgent.

WF-202 — Each farm card shows: status icon and colour, farm name, farm type icon, area, plot or tree count, a one-line plain-language summary of the most significant issue, and the age of the data.

WF-203 — "Updated X ago" refers to the age of the most recent **satellite imagery**, not the last app refresh. Where imagery is older than 7 days, show the reason ("Cloud cover since 28 July").

WF-204 — Farms are sorted by severity first, then by most recently viewed.

WF-205 — Pull to refresh, with a visible refresh button as the non-gesture equivalent.

WF-206 — Empty state: "You have no farms yet" with an *Add your first farm* button leading to A7.

5.2 B2 — Farm detail



WF-207 — The health block lists only measures included in the farm's current plan. Measures outside the plan are shown greyed with a small lock and a tap target that opens the upgrade screen (§9.5) — never hidden entirely. The user should know what they are missing.

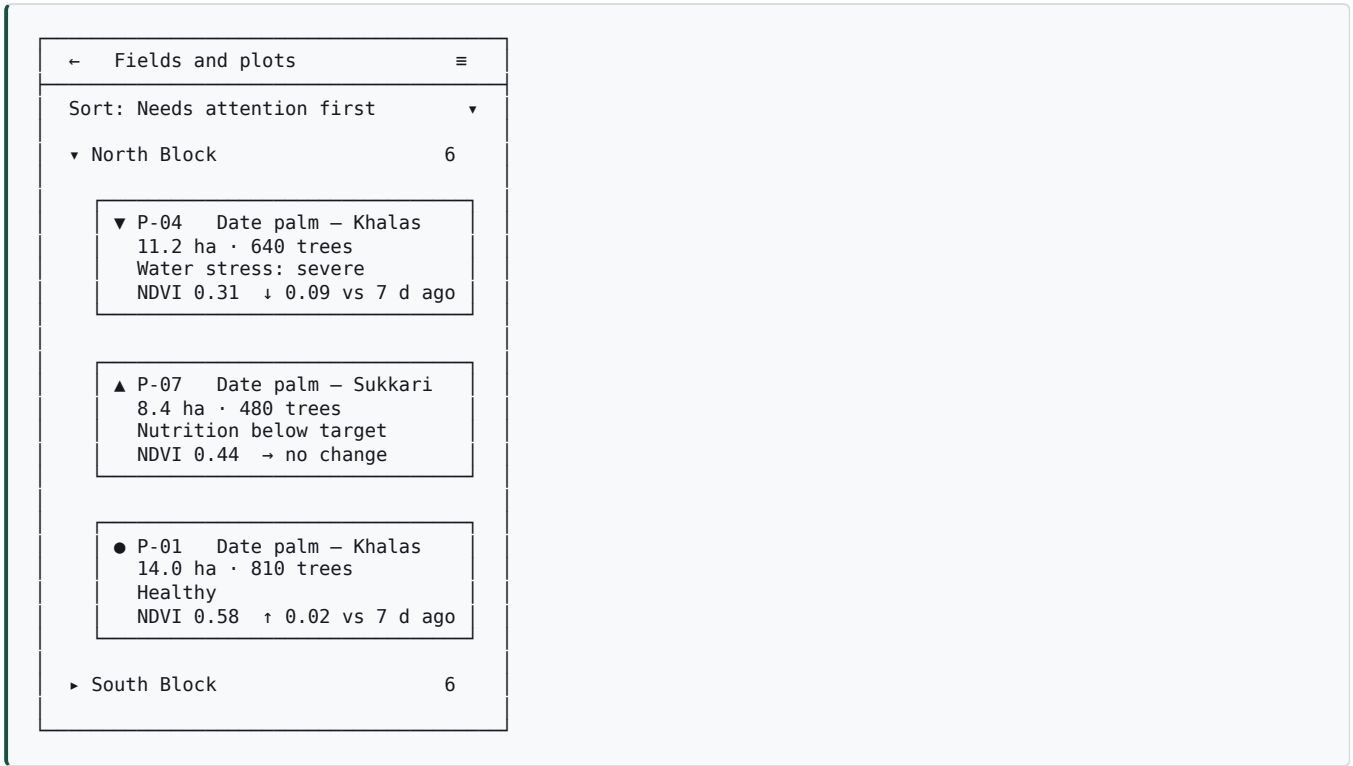
WF-208 — Tapping any row in the health block opens the measure viewer (B7) for that measure across the whole farm.

WF-209 — The *Trees* row is present only for tree and mixed farms.

WF-210 — The weather strip shows current conditions and a 7-day or 14-day forecast according to plan (§9.3), and surfaces any active weather alert inline.

WF-211 — Farm settings (⚙️, top right) is visible to Owner and Supervisor only.

5.3 B3 — Fields and plots

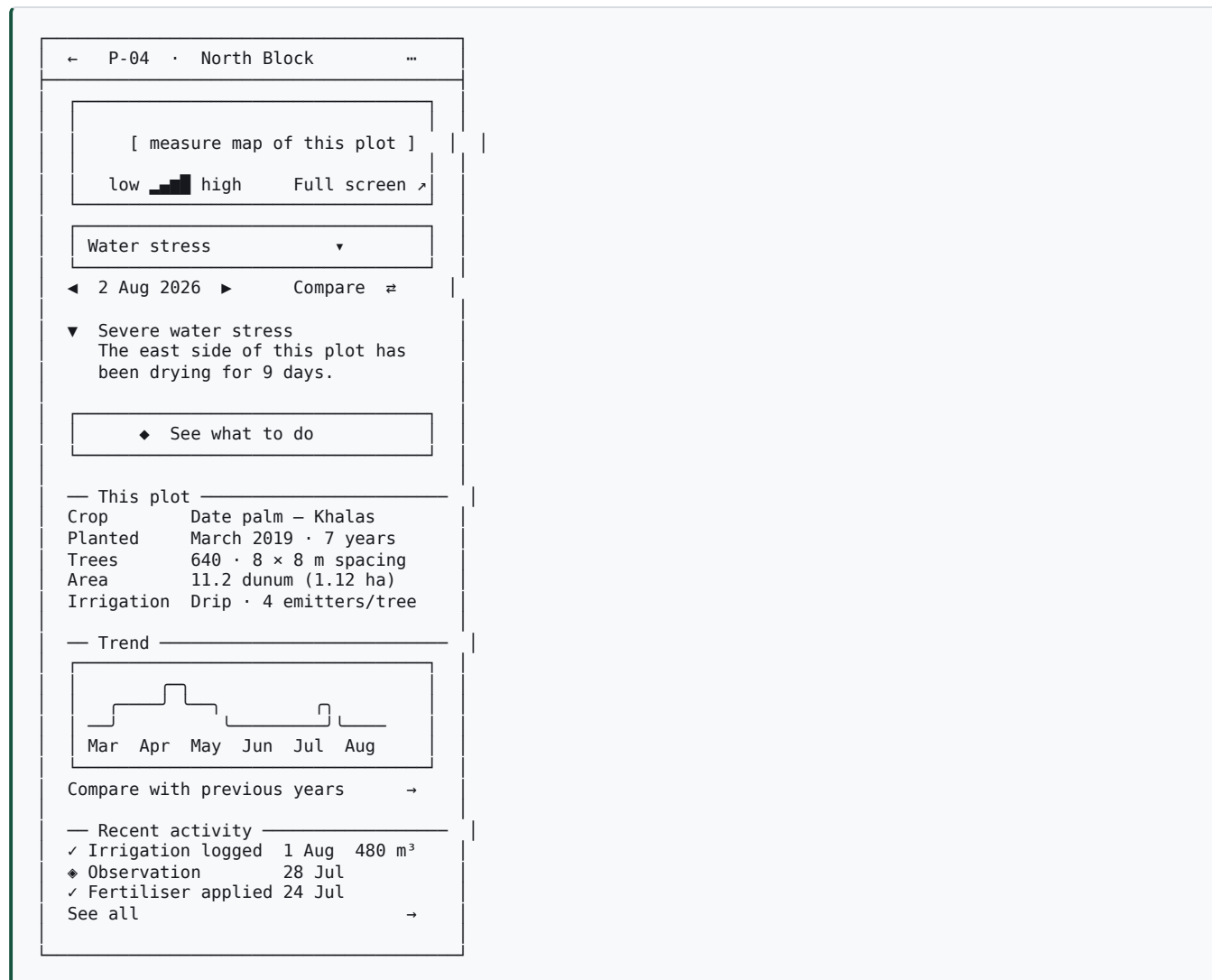


WF-212 — Plots are grouped by block where blocks exist; where a farm has no blocks, the grouping level is hidden entirely rather than showing an empty "Default block". The hierarchy is optional in the interface even though it is always present in the data model.

WF-213 — Sort options: Needs attention first (default), Name, Area, Most recently changed.

WF-214 — Each plot row shows the measure value and its change over the last 7 days, with a direction arrow.

5.4 B4 — Plot detail



WF-215 — The measure selector is a dropdown listing every measure available on this farm's plan, in plain-language names with the technical name secondary:

Shown to user	Technical
Overall health	NDVI
Water stress	NDWI / water-stress measure
Nutrition status	NDRE / chlorophyll
Growth and vigour	EVI
Soil-corrected health	MSAVI
Photosynthesis	as provided by MMC

The plain-language name is what appears everywhere in the interface. The technical name appears in the measure viewer legend and in reports, for the agronomists.

WF-216 — Date stepping with ◀▶ moves between **available imagery dates**, not calendar days. Dates with no imagery are skipped, and the reason is shown if the user reaches the end of available data.

WF-217 — The interpretation line below the map names *where* and *how long*, not only *what*: "the east side has been drying for 9 days", not "severe water stress".

WF-218 — See *what to do* is the primary action and links to the relevant recommendation (D2–D5). If no recommendation exists for this plot, the button is replaced by *Create a task* (E3) pre-filled with the plot.

WF-219 — The  menu contains: Edit plot, Edit boundary, Add crop cycle, Add observation, Create task, Share plot summary, Delete plot. Delete requires typing the plot name to confirm and is Owner-only.

5.4.1 B7 — MEASURE VIEWER, FULL SCREEN

WF-220 — Full-screen measure map, pinch-zoom and pan, with a persistent legend showing the value scale and the units.

WF-221 — Tapping any point on the map shows the value at that point in a small callout, with its coordinates.

WF-222 — The legend scale is fixed per measure, not auto-scaled to the current image, so that week-to-week comparison is valid.

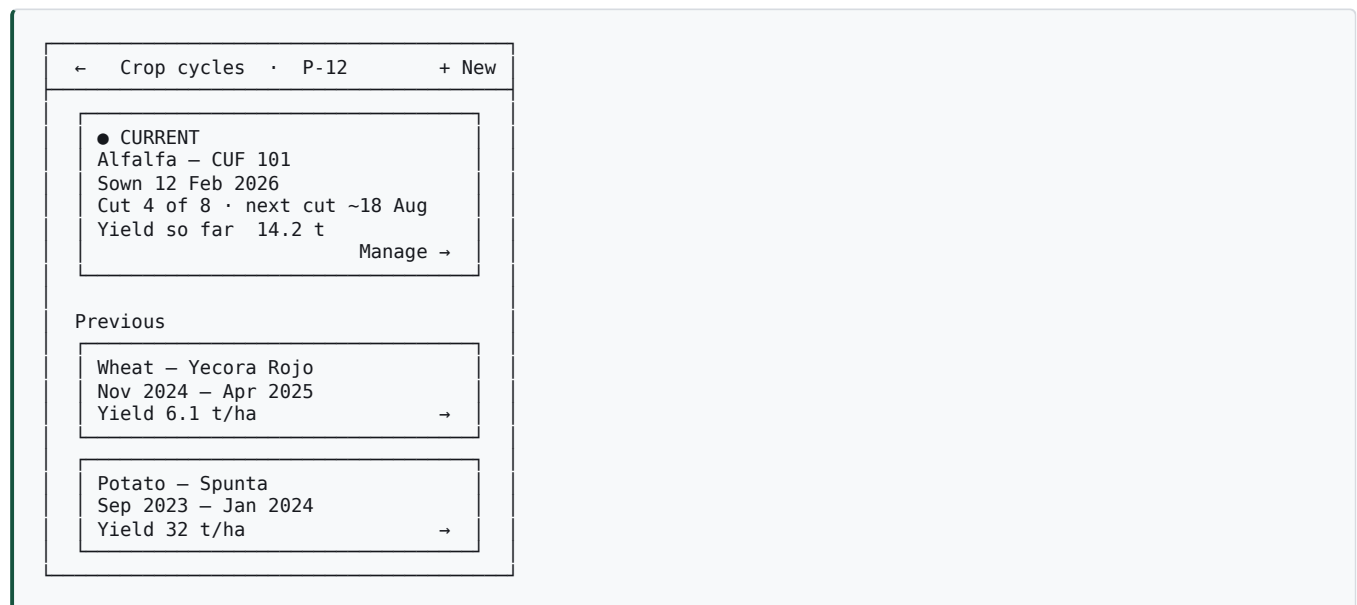
5.4.2 B8 — COMPARE

WF-223 — Side-by-side or swipe-divider comparison of the same plot on two dates, with both dates labelled and the difference in the measure value stated numerically.

WF-224 — A *compare with previous years* mode showing the same calendar week across up to 5 previous years. Advanced crop plan only; on lower plans the control is visible and locked.

5.4.3 B5/B6 — CROP CYCLES

A crop cycle is a first-class record. It is **not** a field on the plot.



WF-225 — Adding a new crop cycle to a plot that already has an open one prompts the user to close the existing cycle first, recording a harvest date and — optionally — a yield. It must not silently overwrite.

WF-226 — Closing a crop cycle never deletes it. The plot's full crop history is retained indefinitely and remains visible. Rotating a field must not erase multi-year history.

WF-227 — Crop cycle fields: crop, variety, sowing/planting date, expected harvest date, actual harvest date, target yield, actual yield, irrigation method, notes. Only crop and start date are mandatory.

WF-228 — The crop picker is searchable, grouped by category (cereals, forage, vegetables, fruit trees, other), and offers the last five crops used on this account at the top.

5.4.4 MULTI-CROP AND HIERARCHY

WF-229 — A farm supports many plots with different crops, varieties and planting dates simultaneously.

WF-230 — The hierarchy Farm → Block → Field → Plot is supported in the data model. In the interface, Block and Field are optional: a user may create plots directly under a farm and never see the intermediate levels.

WF-231 — A user may create, rename, reorder and delete blocks and fields. Deleting a block does not delete its plots; they move up a level, and the user is told so.

5.4.5 INTERCROPPING

WF-232 — A plot may record a **primary crop and one or more secondary crops** (for example date palm with alfalfa beneath). This is a recording capability.

WF-233 — Measure values and health status on an intercropped plot are attributed **per crop**, using the platform's canopy/understory separation. The plot detail screen carries a crop selector alongside the measure selector, defaulting to the primary crop; a *Whole plot* option shows the unseparated reading.

WF-234 — Recommendations are generated per crop within the plot. An irrigation or nutrition recommendation names the crop it applies to, and the task created from it carries that crop.

WF-235 — Where separation cannot be computed for a given date — insufficient resolution, cloud, or an unsupported crop pairing — the app shows the whole-plot reading marked *Combined canopy* and suppresses per-crop recommendations for that date only. It never presents a whole-plot figure as if it were a single crop's.

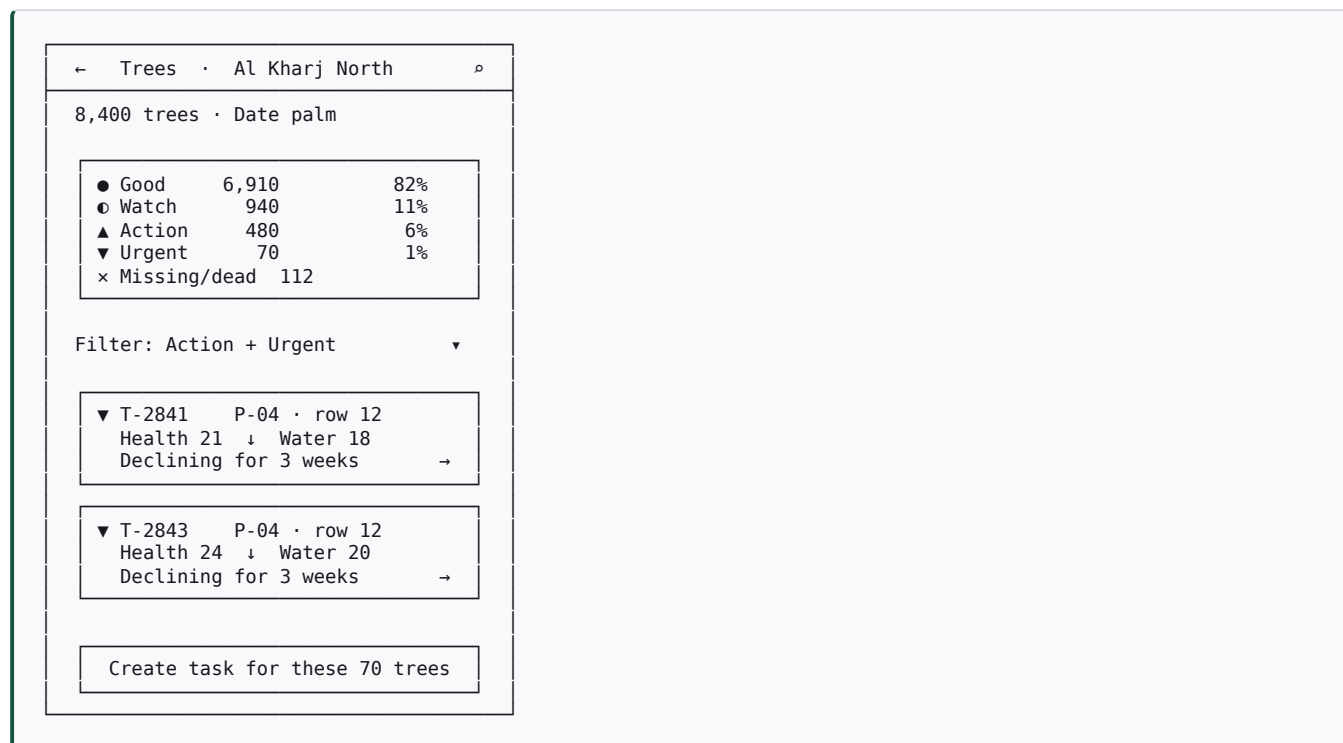
5.4.6 B11 — FARM SETTINGS

WF-236 — Editable: farm name, farm type, address/region, irrigation system, soil type, default language for reports, primary contact. Owner and Supervisor only.

WF-237 — Contains *Transfer ownership* (Owner only, with SMS confirmation) and *Delete farm* (Owner only, requires typing the farm name, warns that all history is lost, and requires a second confirmation after 24 hours for farms with more than 90 days of data).

5.5 B9/B10 — Trees

Shown for tree and mixed farms only. Backed by MMC's tree monitoring platform.



WF-238 — The screen leads with the status distribution; the list below is filtered to what needs action by default.

WF-239 — Filters: status, plot, row, species, planting year, and "declining" (health falling over the last 3 observations).

WF-240 — A bulk action from any filtered list: create one task covering all matching trees.

WF-241 — Tree detail (B10) shows: tree ID, QR code, species and variety, planting year and age, position (plot, row, position, coordinates), current health score and its component measures, a 12-month health trend chart, and the history of observations, tasks and photos attached to that tree.

WF-242 — QR code scanning from the Map tab or the tree list opens that tree's detail screen directly. QR is included from Tree Pro upwards; on Tree Basic the scanner is visible and locked.

WF-243 — Missing and dead trees are shown as a distinct state with their own count, not folded into "urgent".

WF-244 — Where a tree-level measure is not in the client's plan, it is listed and locked rather than omitted (WF-207).

5.6 B13 — Harvest planning and yield

Tree farms on Tree Pro. Four capabilities share one screen: orchard yield estimation, yield optimisation advisory, plot-level harvest planning and the ripeness distribution that drives them.

← Harvest · Al Kharj North

Season 2026 · Date palm

Estimated yield

284 tonnes
± 12% · 33.8 kg/tree

Last season: 261 t ↑ 9%

Ripeness

Ready

Near ready

Developing

Immature

38%

27%

31%

4%

Pick order

1

P-04

Ready 62%

~4 Aug

98 t · 640 trees

→

2

P-01

Ready 41%

~9 Aug

112 t · 810 trees

→

3

P-07

Ready 18%

~17 Aug

74 t · 480 trees

→

◆ Pick P-04 within 5 days. Ripeness is rising fastest on the east side.

Create harvest tasks

WF-245 — Estimated yield is shown for the orchard and per tree, with a confidence range and the previous season for comparison. A figure without a range is never shown.

WF-246 — Ripeness distribution is shown as a proportion of the trees in each stage, for the whole farm and per plot.

WF-247 — The pick order lists plots ranked by readiness, each with its ripe proportion, an estimated pick window, an estimated tonnage and its tree count. Tapping a row opens the plot.

WF-248 — *Create harvest tasks* generates one task per plot in the pick order, pre-filled with the plot, the estimated window as the due date and the tonnage in the description. The supervisor assigns them and saves (§5.10.1).

WF-249 — Yield optimisation advice appears as a single plain-language line above the action, in the same form as every other recommendation: what to do, and why (§5.9).

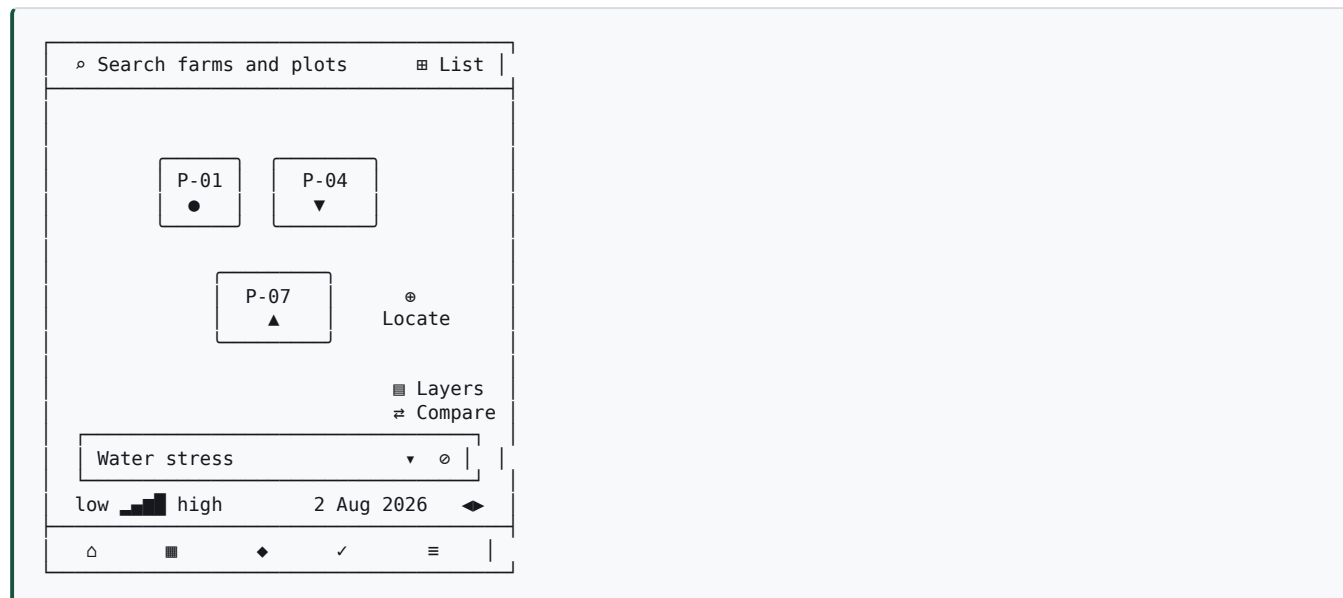
WF-250 — Where the plan does not include harvest planning, the entry point is visible and locked (§9.5).

5.7 Personalised advisory

WF-251 — On Tree Pro, the advice inbox additionally carries orchard-level recommendations generated from the farm's own history — its yield curve, its irrigation record, its recurring stress patterns — rather than from generic thresholds alone. These appear in the same inbox, in the same card format, and are logged like any other recommendation (§6.3).

WF-252 — A personalised recommendation states what it learned from, in one line: "based on three seasons on this farm".

5.8 C1 — Map



WF-253 — The map opens on the user's farms, zoomed to fit all of them, or to the currently selected farm where a farm context is active.

WF-254 — Plot polygons are filled with the currently selected measure and carry their status icon and label at readable size. Labels hide automatically below a zoom threshold rather than overlapping.

WF-255 — Tapping a plot opens a bottom sheet (C3) with: plot name, crop, area, current measure value and change, status line, and two buttons — *Open plot* (B4) and *Create task* (E3). The sheet is draggable to full height without leaving the map.

WF-256 — Layer picker (C2) offers, subject to plan:

- Basemaps: satellite, terrain, street
- Measure layers: all measures available on the plan
- Farm layers: boundaries, block boundaries, plot labels, tree points
- Advanced GIS: soil type, land use, testing labs (Advanced crop plan only)
- VRA layers: sowing, nitrogen, P&K, zoning (per plan)
- Irrigation map (Advanced crop plan; Tree Pro)

WF-257 — Layer selections persist between sessions.

WF-258 — Locked layers appear in the list with a lock icon and open the upgrade screen when tapped.

WF-259 — The user's own GPS position is shown when location permission is granted, with a *Locate me* control that centres on it.

WF-260 — Date stepper on the map moves through available imagery dates for the visible plots, as WF-216.

WF-261 — Compare mode (C4) puts a draggable vertical divider across the map with a different date either side.

WF-262 — Map tiles for the user's own farms are cached for offline use (§11.2). When offline, the map shows cached tiles and a clear banner stating the date of the cached imagery.

WF-263 — Search accepts farm name, plot name, crop name and tree ID.

5.8.1 C5 — BOUNDARY EDITOR

WF-264 — Editing an existing boundary uses the same interaction as A8: tap to add, drag to move, and a visible delete control for the selected vertex.

WF-265 — A boundary change is a **versioned event**, not an overwrite. The previous geometry, the user who changed it and the timestamp are retained and appear in the activity log (F11). Historical analytics remain attached to the geometry that was in force at the time.

WF-266 — Boundary editing requires the `farm.boundary.edit` capability (Owner and Supervisor). It is **not** available offline.

5.9 D1 — Advice inbox

An inbox, not a dashboard: items arrive, are read, are acted on, and are cleared.



WF-267 — Recommendations are grouped by urgency into Today / This week / Later, and within each group sorted by severity.

WF-268 — Every recommendation card states, in this order and in plain language: what to do, how much, and why. The reason is mandatory.

WF-269 — Every recommendation carries two actions inline, without opening the detail screen: **Create task** (E3, pre-filled) and **I did it** (D7).

WF-270 — Filters: farm, plot, type (irrigation, nutrition, protection, weather, harvest), status (needs action / all / done).

WF-271 — A recommendation is marked done when the user records an action against it (D7) or completes a task created from it. It then moves to the Done tab and stops appearing in badges.

WF-272 — Recommendations are never silently deleted or replaced. A superseded recommendation is marked superseded, with a link to the one that replaced it, and remains readable (§6.3).

WF-273 — Where the client's plan does not include advisory, the Advice tab is still present and shows weather alerts (if in plan) plus a locked card describing what would appear here, with an upgrade path. The tab is never empty.

5.9.1 D2 — IRRIGATION ADVICE DETAIL

← Irrigation · P-04 ...

▼ URGENT · issued 3 Aug, 05:00

Apply 62 mm
this week

693 m³ over 11.2 ha
≈ 6 h 20 m at 110 m³/h
≈ 1,083 litres per tree

Suggested split

Mon 4 Aug	21 mm	231 m ³
Wed 6 Aug	21 mm	231 m ³
Fri 8 Aug	20 mm	231 m ³

Why

- Crop water use (ETc) 8.4 mm/day
- Reference ET0 11.2 mm/day
- Crop coefficient Kc 0.75
- Rain forecast this week: none
- Soil moisture at 40 cm: low
- Heat stress expected Tue (44 °C)

Assumptions we used Edit →

Drip · 85% efficiency · sandy loam

Create task

I did it

This is advice, not a prescription.
Check conditions on the ground.

WF-274 — The headline number is shown in three units at once: depth (mm), total volume (m³), and — for tree farms — litres per tree.

WF-275 — Pumping time is calculated and displayed where the plot has a recorded system flow rate. Where it does not, show a prompt to add one rather than hiding the row.

WF-276 — The "Why" section lists every input used, with its value.

WF-277 — *Assumptions we used* is editable. A user who corrects their irrigation efficiency or soil type triggers a recalculation. The edit is recorded against the plot, not just this recommendation.

WF-278 — The disclaimer line is present on every advisory detail screen, in the user's language, and is not dismissible.

5.9.2 D3 — NUTRITION ADVICE DETAIL

WF-279 — Same structure: headline dose, split across application windows, the reasoning, editable assumptions.

WF-280 — Nutrient recommendations are expressed as **elemental N, P, K, Ca, Mg per hectare**, with an optional product translation where the user has recorded which fertiliser products they hold. Never lead with a product name.

WF-281 — Where MMC's platform provides fertiliser insight but not a full fertigation schedule for the client's plan, the screen must say so explicitly rather than producing an unsupported schedule. The combined fertigation scheduler is outside every plan sold and must not be implied anywhere in the app — see §9.3.

5.9.3 D4 — CROP PROTECTION ADVICE DETAIL

WF-282 — Recommendations lead with the **active ingredient and rate**, not a commercial product name.

WF-283 — Product names are drawn from the registered-product register for the user's country — see §6.4.

WF-284 — Every crop protection recommendation displays the **pre-harvest interval** for the user's country, prominently, with the earliest safe harvest date calculated and stated as a date, not as a number of days.

WF-285 — A permanent, non-dismissible notice: *Check the product label and your local regulations before applying. This is advice, not a prescription.*

WF-286 — The screen includes symptom photographs and a short identification guide from MMC's disease directory, so the user can confirm the diagnosis before spraying.

WF-287 — Every crop protection recommendation is written to the advisory log when it is generated — see §6.3.

5.9.4 D6 — WEATHER ALERTS

WF-288 — Alert types: frost, heat stress, high wind (spraying), heavy rain, sandstorm/dust, high humidity (disease). Each states the threshold crossed, the window, and one line on what it means for the farm.

WF-289 — Weather alerts are delivered as push notifications at the criticality defined in §7.2, in the recipient's language.

5.9.5 D7 — RECORD WHAT YOU DID

← What did you do?

Irrigation · P-04 · advised 62 mm

☒ I applied the full amount

☐ I applied a different amount

☐ I did not do this

Choosing a *different amount* reveals a single numeric field with a unit toggle (mm / m³ / litres per tree) and an optional note. Choosing *I did not do this* reveals a short reason list (no water available, pump failure, weather, not needed, other).

WF-290 — Recording an action takes at most **three taps** from the advice card for the common case.

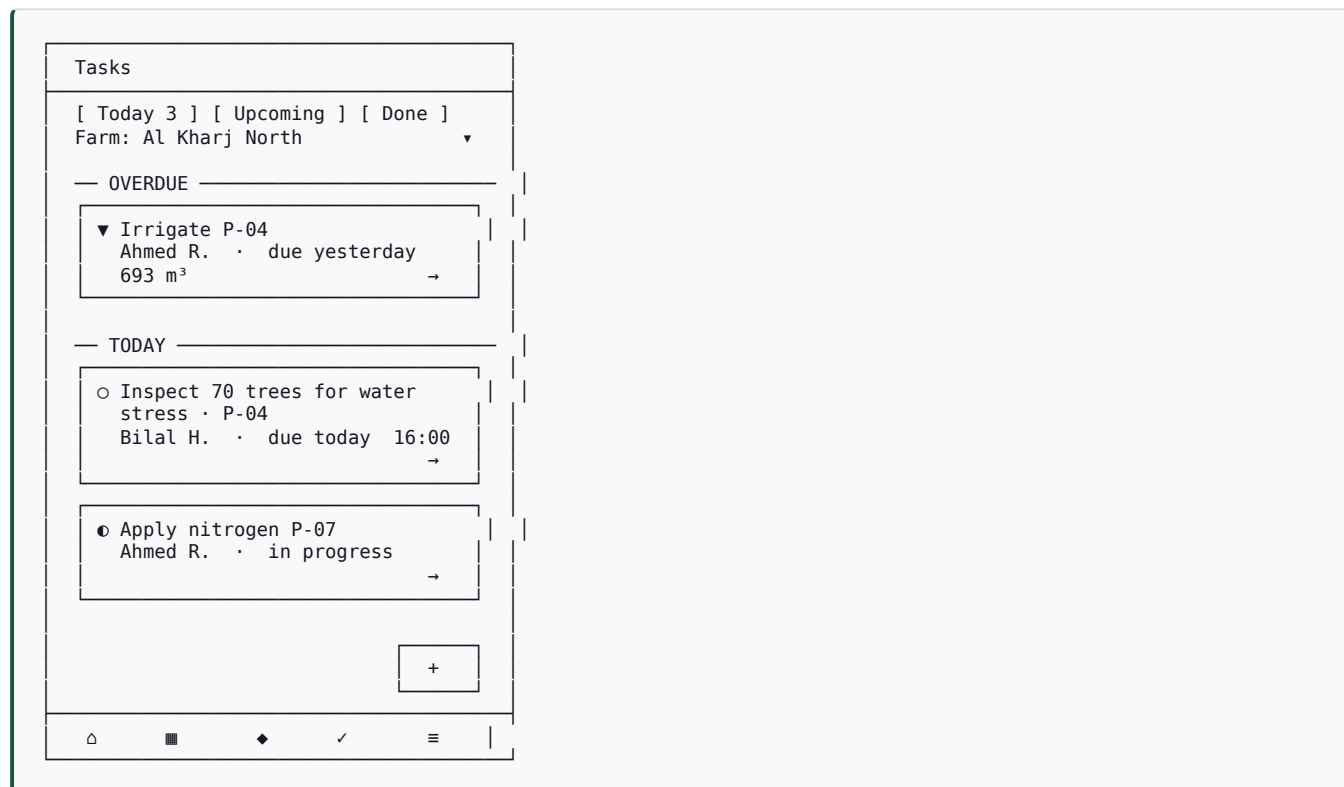
WF-291 — The recorded action writes to the plot's activity history and the farm diary, and is available to reports.

WF-292 — Once actions have been recorded for a plot, the plot detail screen shows an advised-vs-applied comparison over time.

WF-293 — This screen works offline.

5.10 E1 — Tasks

Not covered by the existing feature list. This module must be built.



WF-294 — Task states: Open → In progress → Done, plus Cancelled. No other states, and no approval workflow.

WF-295 — A task has: title, description, type, farm, plot or tree selection, assignee, due date and optional time, priority, and optional attachments.

WF-296 — Task types (used for icons, filtering and reporting): irrigation, fertiliser, spraying, planting, pruning, harvest, inspection/scouting, maintenance, other.

WF-297 — A task may be assigned to **one person**. Team assignment is out of scope; a supervisor who needs three people creates three tasks or one task for a named lead.

WF-298 — A task may cover many plots or many trees — see WF-240.

WF-299 — Overdue tasks are always shown first, always visually distinct.

5.10.1 E3 — CREATING A TASK

WF-300 — Creating a task from a recommendation pre-fills title, description (including what and how much), type, plot and a suggested due date. The supervisor adds an assignee and saves.

WF-301 — The assignee picker lists only users who have access to the selected farm, shows each person's role, and shows how many open tasks they already have.

WF-302 — On save, the assignee is notified through their configured channel (§7) in **their** language, not the creator's.

5.10.2 E2/E4 — TASK DETAIL AND COMPLETION, WORKER VIEW

←

↑

Irrigate plot P-04

693 cubic metres
About 6 hours 20 minutes

Today, before 18:00

■ Show me where

✓ Mark as done

I cannot do this

WF-303 — The Worker view of a task is stripped to: what, how much, where, when, and one big button. No status chips, no metadata, no comment threads.

WF-304 — *Show me where* opens the map centred on the target plot or tree, with a line and a distance from the worker's current position. Not a routing engine.

WF-305 — *Mark as done* opens E4:

← Mark as done

Take a photo
(optional)

How much did you apply? (optional)

680 m³ ▼

Note (optional)

Confirm

● Offline – this will send when you have signal

WF-306 — Everything on the completion screen is optional except the confirm tap. A worker who cannot type must still be able to complete a task.

WF-307 — Photographs are automatically stamped with GPS coordinates, timestamp, task ID, plot ID and user ID. See §A.3.

WF-308 — The whole completion flow works offline, confirms locally and immediately, and syncs later (§11).

WF-309 — *I cannot do this* records a reason from a short picture-supported list and notifies the task creator immediately.

5.10.3 E6 — FIELD OBSERVATION / SCOUTING

WF-310 — A standalone observation can be captured from the Tasks tab, the map, a plot or a tree, without an existing task. Fields: photo(s), category (pest, disease, weed, water, damage, other), free note, severity, automatic location.

WF-311 — Observations appear on the map as pins, on the plot's activity history, and in the farm diary.

WF-312 — Where the plan includes photo-based disease detection (E7), photographing a leaf returns an automated identification with a confidence level, which the user may accept or reject. An automated identification is never presented as certain.

WF-313 — Works fully offline. Photo disease detection requires a connection and is queued with a clear "will be checked when you have signal" state.

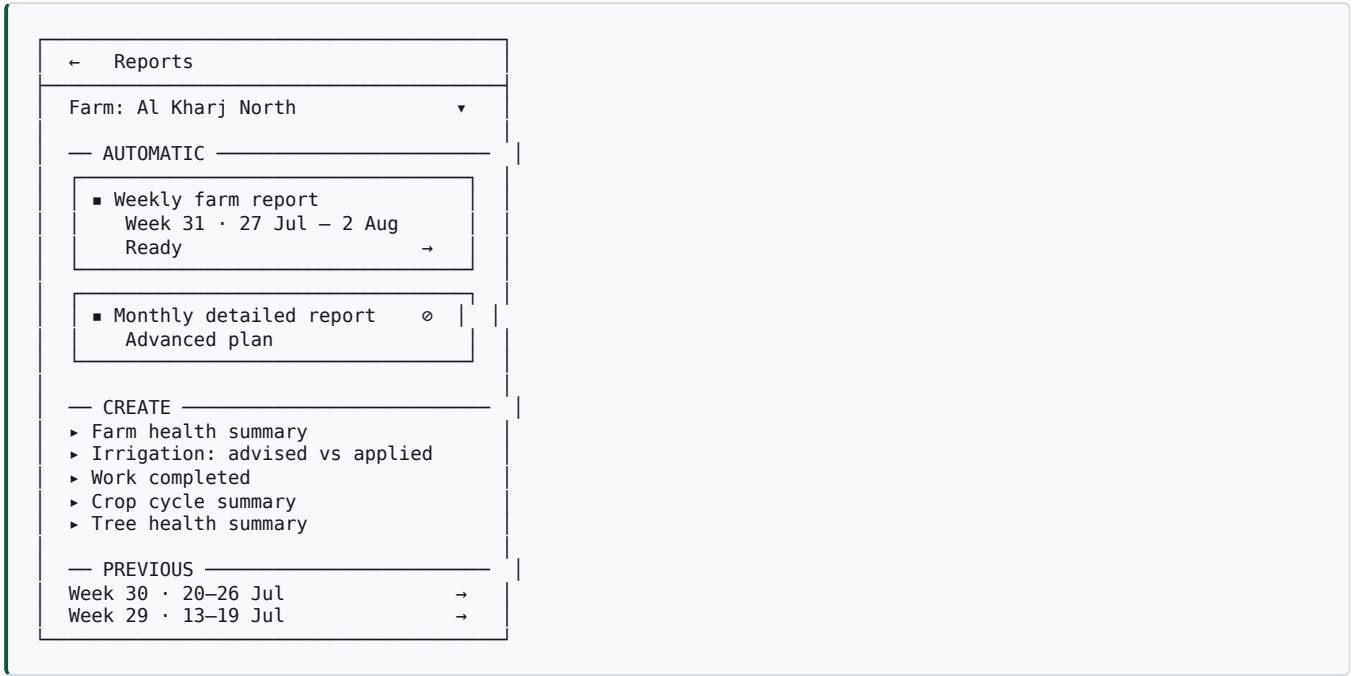
5.11 F — The More tab



WF-314 — Menu contents are filtered by role. A Worker sees only: profile, language, notifications, data and storage, help, contact Wafra, version, log out.

WF-315 — The version and build number are always visible on this screen.

5.11.1 F1 — REPORTS



WF-316 — Reports are generated **server-side as PDF**, never assembled on the device. Excel export is offered for the tabular reports.

WF-317 — Every report is produced in the language the user requests, from the same string catalogue as the app, and carries the Wafra brand only.

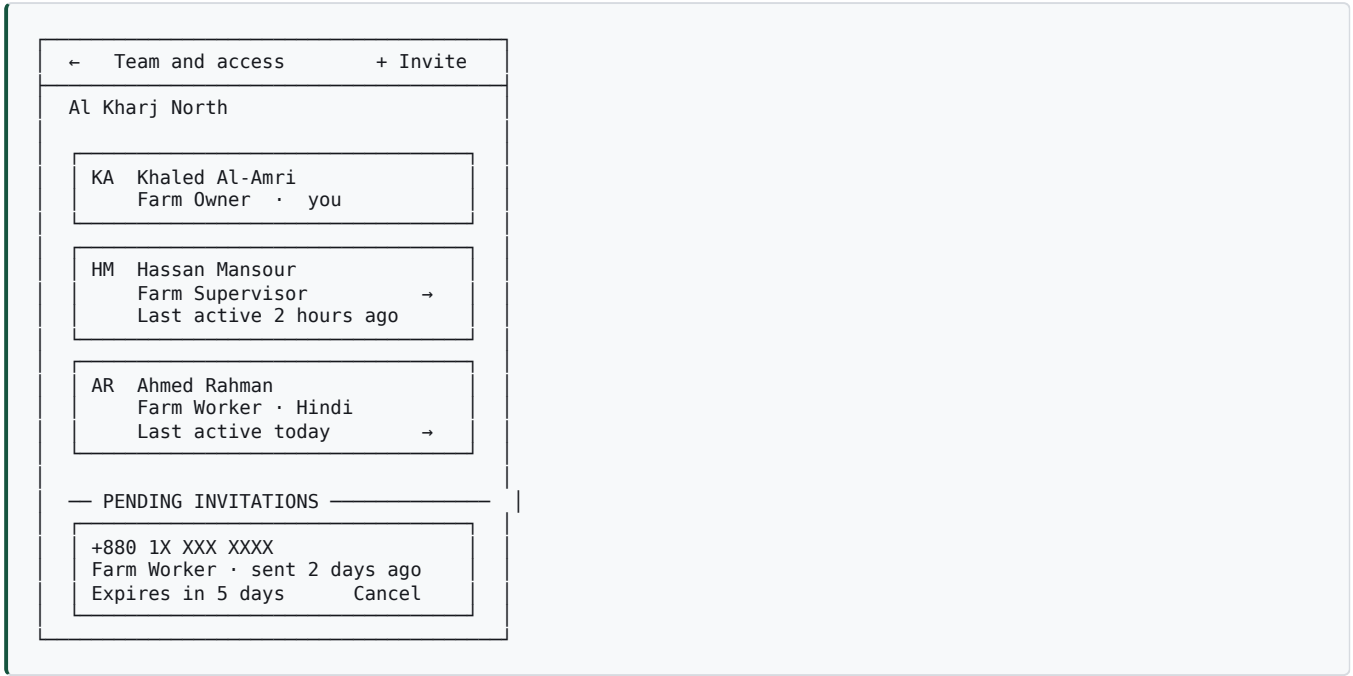
WF-318 — Reports are shareable through the OS share sheet (WhatsApp, email, file save).

WF-319 — The **Irrigation: advised vs applied** report is not in the existing report set and must be built. Per plot per period: advised volume, applied volume, difference, and the resulting health trend.

WF-320 — Reports outside the client's plan appear locked, not hidden.

WF-321 — The weekly report is also delivered by email to any user with an email address and by WhatsApp where the user has opted in.

5.11.2 F2-F4 — TEAM AND ACCESS



- WF-322** — Pending invitations are listed with their expiry and a one-tap cancel. An owner must always be able to see who has been invited and revoke it.
- WF-323** — Each member row shows their role and their app language, so the assigner knows which language a task will be delivered in.
- WF-324** — Invite (F3) is by mobile number (primary), email, link, 6-character code or QR — see §4.1. Role is selected during the invite and cannot be left blank.
- WF-325** — Invitations may be scoped to specific farms, not necessarily the whole account.
- WF-326** — QR-code invitations produce single-use, short-expiry, role-scoped tokens.
- WF-327** — Removing a member takes effect immediately, invalidates their active sessions for that farm, and is recorded in the activity log. Their historical task completions and observations are **retained and remain attributed** to them.

5.11.3 F5/F6 — SUBSCRIPTION



WF-328 — Crop and tree subscriptions are **separate** and displayed separately. A client may hold one, the other, or both, at different levels.

WF-329 — Trial status shows days remaining, prominently, from day one.

WF-330 — Purchase and renewal are handled through **Apple In-App Purchase and Google Play Billing**. The app must not attempt to take payment by any other means inside the app, and must not link to an external purchase page for the same product, as this violates both stores' rules and will fail review.

WF-331 — Entitlement is **always determined by the server**, never by the local receipt alone. The client validates the store receipt with the backend, the backend records the entitlement, and the app asks the backend what the user may do. See §9.5.

WF-332 — The *Contact Wafra* route for invoicing and annual contracts is informational only — it opens the contact screen (F13). It must not describe prices or take payment, or the app will be rejected.

5.11.4 F7–F10 — SETTINGS

WF-333 — Language and region (F8): app language, area unit preference (dunum/hectare/acre — dunum default across the region), water unit (m³ / litres / mm), temperature (always Celsius), date calendar (Gregorian / Hijri / both), and currency display.

WF-334 — Notifications (F9): per-channel and per-category toggles, quiet hours, and a channel preference for tasks (push / WhatsApp / SMS). See §7.

WF-335 — Data and storage (F10): current cache size, "download imagery on Wi-Fi only" toggle, "clear cached imagery" action, and a **pending sync list** showing exactly what has not yet uploaded with a manual *Sync now* button.

WF-336 — A **shared device** toggle (WF-161).

WF-337 — Settings includes *Delete my account*, which explains what will be deleted and what will be retained, requires confirmation, and creates a deletion request processed within 30 days. This is required by both app stores and by regional data protection law.

5.11.5 F11 — ACTIVITY LOG

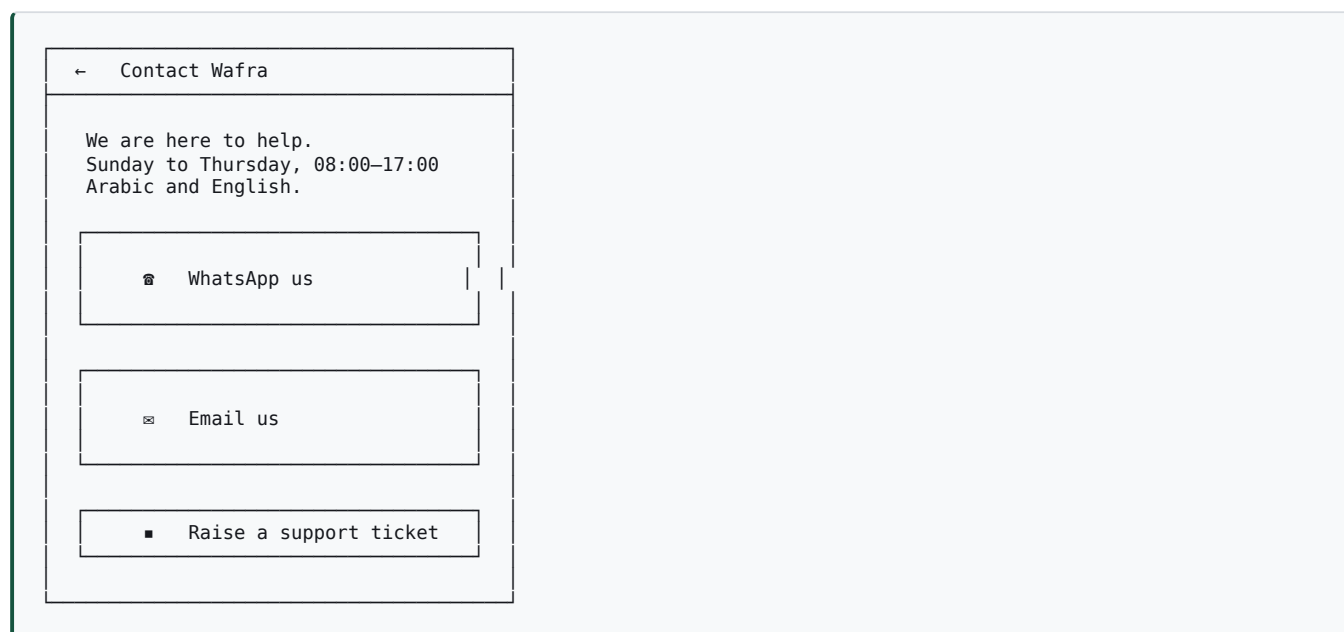
WF-338 — Owner-only. A chronological, filterable, read-only record of: boundary changes, crop cycle changes, task creation and completion, input logging, member additions and removals, role changes, subscription changes, and advice acknowledgements. Each entry records who, what, when and from where (farm context).

WF-339 — The activity log is **append-only**. No user, including an Owner, can edit or delete entries.

5.11.6 F12/F13 — HELP AND CONTACT WAFRA

WF-340 — Help contains a searchable user guide, in all five languages, with short illustrated articles. The guide is served from the backend so it can be updated without an app release.

WF-341 — Contact Wafra presents exactly two large buttons plus a support ticket option:



WF-342 — WhatsApp opens a chat with **+966 54 810 0443**, pre-filled with a short message containing the user's account reference, app version and current screen.

WF-343 — Email opens the device mail client to **info@wafragreen.com**, with the same diagnostic details in the body.

WF-344 — Both destinations are configuration values fetched from the backend, not compiled into the app, so they can change without an app release.

WF-345 — The support ticket route uses the existing ticketing capability where the plan includes it. Tickets are routed to Wafra as first line.

6 Advisory layer

Every recommendation the app shows is produced by the advisory layer, which MMC builds and operates. This chapter states what that layer must produce, record and withhold. It does not specify how the models work.

The screens that present advice are in §5.9. This chapter covers what sits behind them.

6.1 What the platform already provides

Taken from the crop and tree feature lists. Plan names are the Wafra names (§9).

Crop

Advisory capability	From plan
Crop disease directory	Basic
Operations advisory	Pro
Expert opinion request	Pro
Disease forecasting	Pro
Fertiliser insights	Pro
ET / ET0 / GDD data	Pro
Irrigation scheduler	Advanced
Photo-based disease detection	Advanced
Growth stage modelling	Advanced
Farm anomaly detection	Advanced
Custom weather alerts	Advanced
Connect with Agro Doctor	Advanced
VRA maps (sowing, nitrogen, P&K)	Advanced

Trees

Advisory capability	From plan
Tree disease directory	Basic
Tree / orchard planner (seasonal calendar)	Basic
Satellite-based disease forecasting	Pro
Pest and disease alert	Pro
Early stress warning	Pro
Stress detection	Pro
Ripeness stage distribution	Pro

Advisory capability	From plan
Progressive harvest alert thresholds	Pro
Harvest alert queue	Pro
Tree irrigation schedule	Pro
Irrigation efficiency rating	Pro
Connect with Agro Doctor	Pro

Two capabilities sit above every plan sold and must not be surfaced: the combined **fertigation scheduler** (the tier behind Wafra Advanced provides irrigation scheduling only) and **personalised AI advisory on trees**, which sits above Tree Pro.

6.2 What must be built

Nothing in the existing feature lists covers the following. Each is specified below.

Why it does not exist today	
Immutable advisory log	No equivalent in the feature lists
Registered product register, per country	The disease directories describe diseases, not registered products, rates or pre-harvest intervals
Advised-versus-applied record	No equivalent
Per-crop attribution on intercropped plots	Not addressed
Suppression and fallback rules	Not addressed

6.3 The advisory log

WF-600 — Every recommendation is written to an advisory log at the moment it is generated, before it is delivered to any user. The record contains: the rendered advice text, the language it was rendered in, the rule version, a snapshot of every input value used, the plot or trees it applies to, the crop it applies to, the recipients, the delivery channels and the timestamp.

WF-601 — The advisory log is **write-once**. No update or delete path exists at any level, including database administration procedures. It is stored separately from the operational database, with its own backup schedule.

WF-602 — Retention is **7 years**. Account deletion does not remove advisory log entries; it removes the personal identifiers attached to them.

WF-603 — A superseded recommendation is marked superseded and linked to the one that replaced it. Neither record is deleted, and both remain readable in the app (\$5.9).

WF-604 — The log is exportable, on request, for a named client and date range, in a documented machine-readable format.

WF-605 — Acknowledgement and action are recorded against the log entry: who saw the advice, when, and what they recorded doing about it (\$5.9.5).

6.4 Registered product register

This specification assumes MMC owns, sources and maintains the registered-product register for every country the app is sold in, and carries responsibility for keeping it current. Whether that is the right allocation is an open item for Wafra to settle internally — see §14.

WF-606 — A register of plant protection products, held per country, containing at minimum: active ingredient, registered product name, registration number, registrant, approved crops, approved rate range and unit, pre-harvest interval in days, re-entry interval, and the date the entry was last verified.

WF-607 — A product is only shown to a user whose farm is in a country where that product is registered. A product not registered in that country is never displayed, in any language, in any screen or report.

WF-608 — Recommendations lead with the **active ingredient and rate**. Product names appear only where the register holds a verified registration for that country.

WF-609 — The pre-harvest interval is taken from the register for the user's country, and the earliest safe harvest date is calculated and presented as a date, not as a number of days (§5.9.3).

WF-610 — Register entries are versioned with an effective-from date. A recommendation records which version it used, so past advice can be reconstructed against the rules that applied at the time.

WF-611 — The register is updated server-side and takes effect without an app release.

WF-612 — Every change to the register is recorded with who changed it, when, and what changed.

WF-613 — Where an entry has not been verified within 12 months, product names from that entry are withheld and the recommendation falls back to active ingredient and rate only.

6.5 Rule versions and reproducibility

WF-614 — Every recommendation carries a `rule_version`. A given rule version, given the same inputs, produces the same recommendation.

WF-615 — Every input used is stored with the recommendation, with its value, unit and source. This is what populates the *Why* section on the advice screens (§5.9) and what makes a past recommendation reconstructable.

WF-616 — Assumptions a user can correct — irrigation efficiency, soil type, system flow rate — are stored against the plot, not against the recommendation, and a correction triggers recalculation from the next generation onwards. The superseded recommendation is retained (WF-603).

6.6 When advice is withheld

WF-617 — Where the client's plan does not include a given advisory capability, no recommendation of that type is generated. The app shows a locked card rather than an empty tab (§5.9).

WF-618 — Where canopy separation cannot be computed for an intercropped plot on a given date, per-crop recommendations are suppressed for that date and the whole-plot reading is shown marked *Combined canopy* (§5.4.5).

WF-619 — Where a mandatory input is missing, no recommendation is generated with a substituted value. The app states which input is missing and offers the screen that collects it.

WF-620 — Crop protection recommendations above a defined risk threshold — rate at the top of the approved range, a pre-harvest interval inside the expected harvest window, or a product flagged as restricted — are held for agronomist

review before delivery. The threshold is configuration, not code.

6.7 Liability controls

WF-621 — Every advisory detail screen carries a non-dismissible line stating that the recommendation is decision support and not a prescription, in the user's language.

WF-622 — Every crop protection recommendation carries a non-dismissible instruction to check the product label and local regulations before applying.

WF-623 — The Terms of Use state that recommendations are advisory, that the farmer remains responsible for what is applied, and that legal compliance with local pesticide regulation rests with the applicator. Accepted at registration (§4.5).

WF-624 — Advice is never presented as certain where it is not. Automated disease identification carries a confidence level and can be rejected by the user; forecasts state the window they cover.

7 Notifications

7.1 Channels

WF-650 — Three channels: **push notification**, **WhatsApp**, and **SMS**.

WF-651 — Every notification is sent in the **recipient's** saved language, not the sender's, not the account's. This is a hard dependency and it affects the backend, not just the app.

WF-652 — WhatsApp delivery uses the WhatsApp Business API with pre-approved message templates. Approval takes time and must be started in week 1. Templates required at launch: task assigned, task overdue, urgent advice, weather alert, weekly report ready, invitation.

WF-653 — SMS is the fallback where WhatsApp delivery fails or the recipient has not opted in.

7.2 Categories and defaults

Category	Default channel	Can user disable?
Task assigned to me	Push + WhatsApp	No (can change channel)
Task overdue	Push	Yes
Urgent advice	Push + WhatsApp	No
Action-needed advice	Push	Yes
Weather alert — severe	Push	No
Weather alert — routine	Push	Yes
Weekly report ready	Push + email	Yes
Team member joined	Push	Yes
Trial ending	Push + email	Yes
Marketing	Off by default	Yes

WF-654 — Categories marked "no" above are safety- or contract-critical and may not be switched off entirely, though the user may change channel.

WF-655 — Quiet hours are user-configurable, default 21:00–05:00 local, and apply to every category except severe weather alerts and urgent advice.

WF-656 — Every notification carries a deep link that opens the exact object it concerns — task, recommendation, plot, farm, report or invitation — from a cold start, and after login where the user is signed out.

WF-657 — Notification volume is capped at 6 per user per day, excluding task assignments and severe weather. Beyond the cap, items are batched into a single digest.

8 Roles and permissions

8.1 The model

WF-670 — Permissions are implemented as a **capability matrix**. Three named role templates ship at launch, each being a named set of capabilities. Adding a fourth role later must not require re-architecting.

WF-671 — Capabilities are the unit of enforcement, not role names. Application code checks `can(user, 'task.assign', farm)`, never `if (user.role === 'supervisor')`.

8.2 Capability matrix

Capability	Owner	Supervisor	Worker
farm.view	✓	✓	✓ (assigned farms)
farm.create	✓	—	—
farm.edit	✓	✓	—
farm.delete	✓	—	—
farm.transfer	✓	—	—
farm.boundary.edit	✓	✓	—
plot.create	✓	✓	—
plot.delete	✓	—	—
croppcycle.manage	✓	✓	—
analytics.view	✓	✓	—
advice.view	✓	✓	—
advice.acknowledge	✓	✓	—
input.log	✓	✓	✓ (own tasks only)
task.view.all	✓	✓	—
task.view.own	✓	✓	✓
task.create	✓	✓	—
task.assign	✓	✓	—
task.complete	✓	✓	✓ (own only)
observation.create	✓	✓	✓
tree.view	✓	✓	✓ (assigned)
tree.override	✓	✓	—

Capability	Owner	Supervisor	Worker
report.view	✓	✓	—
report.export	✓	✓	—
member.invite	✓	✓ (Worker only)	—
member.remove	✓	—	—
member.role.change	✓	—	—
subscription.view	✓	—	—
subscription.manage	✓	—	—
auditlog.view	✓	—	—

WF-672 — A Supervisor may invite Workers but not other Supervisors or Owners.

WF-673 — Access is granted **per farm**, not per account. A user may be a Supervisor on one farm and have no access to another under the same Owner.

WF-674 — A farm may have more than one Owner. Ownership transfer and the addition of a second Owner are explicit, SMS-confirmed actions.

8.3 Enforcement

WF-675 — Every capability is enforced server-side. Every endpoint independently verifies the caller's capability for the specific object being acted on.

WF-676 — Session tokens carry the user identity only. Role and entitlement are resolved server-side on every request, never trusted from the token payload, so that a role change or a cancelled subscription takes effect immediately rather than at next login.

8.4 Invitations

WF-677 — Invitation tokens are single use, expire after 7 days, are bound to the role and farm scope chosen at creation, and are revocable at any time by the Owner.

WF-678 — An invitation link that has been used, expired or revoked shows a clear message and a *Contact the farm owner* action. It never fails silently and it never grants partial access.

9 Plans and entitlements

9.1 The plan structure

WF-700 — Two plan families, and a combined family for clients holding both:

Family	Plans, in order	Offered when
Crop	Basic → Pro → Advanced	The account has crop farms only
Trees	Basic → Pro	The account has tree farms only
Complete	Basic → Pro → Advanced	The account has both, or any mixed farm

WF-701 — Plan names in the interface are exactly **Basic**, **Pro**, **Advanced**. Pro sits between Basic and Advanced. Any supplier-side documentation that orders these differently must not leak into the app.

WF-702 — The mapping from a plan to the underlying platform tier is a **server-side configuration**, not a hard-coded assumption in the app:

Plan	Underlying tier
Crop Basic	Crop "Basic"
Crop Pro	Crop "Advanced"
Crop Advanced	Crop "Professional"
Tree Basic	Tree "Basic"
Tree Pro	Tree "Advanced" — see §9.4
Complete Basic	Crop "Basic" + Tree "Basic"
Complete Pro	Crop "Advanced" + Tree "Advanced"
Complete Advanced	Crop "Professional" + Tree "Advanced"

9.1.1 COMBINED HOLDINGS

WF-703 — An account holds **one plan at a time**, whatever mix of farms it contains. Two concurrent subscriptions are never created.

WF-704 — The plan family offered is derived from the account's farms, not chosen by the user: crop farms only → Crop family; tree farms only → Trees family; anything else → Complete.

WF-705 — Adding a farm of the other type to a single-family account puts that farm in a **pending** state: it is visible, its boundary is kept, and its analytics are replaced by the upgrade path to the matching Complete plan. Nothing already paid for is affected.

WF-706 — Archiving every farm of one type offers a downgrade from Complete to the remaining single family, effective at the next renewal.

WF-707 — Trees are counted for entitlement at the account level across all tree farms, and crop area likewise. A Complete plan resolves to one feature-key set covering both.

9.2 Trial

WF-708 — Every new account receives **one free month** on the plan they select at A12. There is no permanently free tier.

WF-709 — Trial state is held server-side. Reinstalling the app, or signing up again on the same mobile number, does not produce a second trial.

WF-710 — Trial expiry warnings at 7 days, 3 days and 1 day remaining, by push and email.

WF-711 — On trial expiry without a subscription, the account enters read-only state: the user can still see their farms, boundaries, history and past reports, but gets no new analytics, no advice and no task creation. Data is retained for 12 months before deletion, and the user is told so explicitly.

9.3 Feature availability — crop plans

Derived from MMC's crop feature list. Wafra plan names in bold.

Capability	Basic	Pro	Advanced
Satellite resolution	10 m	10 m, 3 m+	10 m, 3 m, 1 m
Base satellites (S1, S5)	✓	✓	✓
Higher-resolution satellites (S6)	—	—	✓
Cloud-free data	—	—	✓
Add farms	✓	✓	✓
Farm listing and search	—	✓	✓
Cadastral import	—	—	✓
Current weather + maps	✓	✓	✓
Weather forecast	7 days	14 days	14 days
Historical weather	—	✓	✓
Custom weather alerts	—	—	✓
ET / ET0 / GDD data	—	✓	✓
Advanced GIS layers	—	—	✓
Farm diary	✓	✓	✓
Dashboard	basic	advanced	advanced
Scouting	✓	✓	✓
1-year progress graph	✓	✓	✓
Weekly farm report	✓	✓	✓
Guaranteed report delivery time	—	✓	✓
Growth stage modelling	—	—	✓
Detailed monthly report	—	—	✓

Capability	Basic	Pro	Advanced
Anomaly detection	—	—	✓
Compare last 5 years	—	—	✓
Soil moisture / temperature	root zone	to 1 m	to 3 m
Fertiliser insights	—	✓	✓
Irrigation scheduler	—	—	✓
Disease directory	✓	✓	✓
Disease forecasting	—	✓	✓
Photo-based disease detection	—	—	✓
Basic maps	✓	✓	✓
Advanced / topographic / compare maps	—	✓	✓
VRA maps (sowing, N, P&K)	—	—	✓
Irrigation map	—	—	✓
Operations advisory	—	✓	✓
Expert opinion request	—	✓	✓
Agro Doctor	—	—	✓
Time series viewer	—	✓	✓
User guide, support tickets	✓	✓	✓
Multi-user access and roles	✓	✓	✓
Five languages	✓	✓	✓
Tasks	✓	✓	✓
Offline capture	✓	✓	✓
Contact Wafra	✓	✓	✓

The five rows in bold at the bottom are Wafra requirements available on every plan. They are not tier differentiators — they are what makes the product usable.

The combined fertigation scheduler and NPK reports sit above the tier Crop Advanced maps to today. Whether to bring them in, and at what commercial cost, is an open item — see the open items. Until it is settled the app does not present a fertigation schedule (WF-170, WF-281).

9.4 Feature availability — tree plans

Capability	Basic	Pro
Individual tree mapping and crown detection	✓	✓
Tree count per block / plot	✓	✓

Capability	Basic	Pro
Tree species identification	—	✓
Planting gap detection	—	✓
Plant density mapping	—	✓
Add orchards / plots by drawing	✓	✓
Nutrition status, overall health, water stress	✓	✓
Fruit-tree health, photosynthesis efficiency	—	✓
Early stress warning	—	✓
Chlorophyll / nutrient content	—	✓
Water availability, stress detection	—	✓
Base satellites (S1, S5)	✓	✓
Cloud-free (SAR) data	—	✓
Tree disease directory	✓	✓
Satellite disease forecasting	—	✓
Ripeness stage distribution	—	✓
Progressive harvest alert thresholds	—	✓
Average canopy density and vigour	✓	✓
Average chlorophyll, canopy water content	—	✓
Stress status, tree health score, canopy vigour	✓	✓
Chlorophyll, water content per tree	—	✓
Tree list and per-tree card	✓	✓
QR code per tree	—	✓
VRA maps — planting, nitrogen	—	✓
Current weather, 7-day forecast	✓	✓
15-day forecast	—	✓
ET / ET0	—	✓
Tree irrigation schedule	—	✓
Irrigation efficiency rating	—	✓
Seasonal orchard planner	✓	✓
Compare plots over time	—	✓
Agro Doctor	—	✓
Harvest alert queue	—	✓
Pest and disease alerts	—	✓

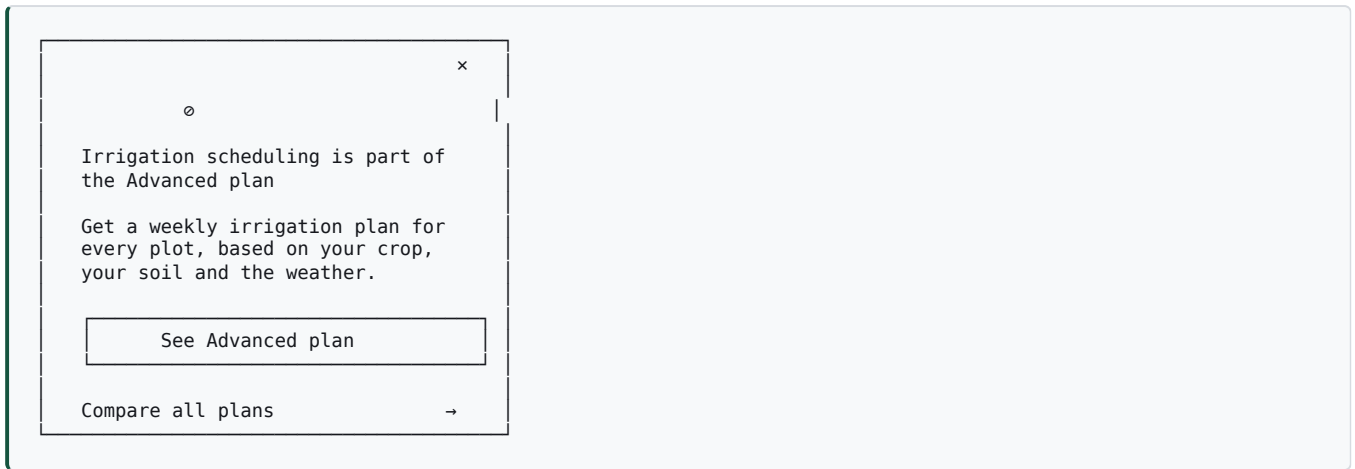
Capability	Basic	Pro
Tree and plot dashboards	✓	✓
Health dashboard and alerts	—	✓
Weekly orchard report	—	✓
User guide	✓	✓
Support tickets	—	✓
Higher-resolution satellites (S6, S7)	—	✓
Soil-corrected health, modified soil health	—	✓
Photosynthetic activity, dense canopy, water infrastructure	—	✓
Canopy structure per tree	—	✓
Yield estimation (orchard level)	—	✓
Orchard yield optimisation advisory	—	✓
Plot-level harvest planning	—	✓
Personalised AI advisory	—	✓
Scouting tasks and field visit workflows	—	✓
Hyper-local plot weather	—	✓
Custom weather alerts	—	✓
VRA maps — P & K, zoning	—	✓
Detailed monthly report	—	✓
Multi-user, languages, tasks, offline, contact	✓	✓

The twelve capabilities from *higher-resolution satellites* downwards sit in the supplier's **Advanced** tree tier, above the tier Tree Pro maps to today. Including them in Tree Pro is a change to the commercial arrangement — see the open items. The screens for the four that are user-facing are specified in §5.6.

9.5 How locked features behave

WF-712 — A feature outside the user's plan is **visible and locked**, never hidden, unless it would be meaningless (for example, tree screens for a crop-only client).

WF-713 — A locked feature shows a lock icon in place of its value or content. Tapping it opens a single, consistent upgrade sheet:



WF-714 — The upgrade sheet names the plan required and describes the benefit in one sentence of plain language. It never shows a technical feature name.

WF-715 — Entitlement checks happen server-side on every request. A modified client must not be able to obtain data outside its plan.

10 Internationalisation (i18n)

10.1 Languages

WF-750 — Five languages ship at launch: **English, Arabic, Hindi, Bengali, Pashto**.

WF-751 — **Arabic and Pashto are both right-to-left**. Pashto is written in a modified Perso-Arabic script and requires full RTL treatment. Treating only Arabic as RTL is a defect.

WF-752 — RTL treatment includes: layout mirroring, navigation and back-arrow direction, list chevrons, progress and slider direction, chart axis direction, icon mirroring where an icon implies direction, and correct bidirectional handling of mixed Latin/Arabic strings such as "P-04" inside an Arabic sentence.

WF-753 — Numbers use **Western Arabic numerals (0–9)** in all languages including Arabic, unless the user chooses Eastern Arabic numerals (٠-٩) in Settings. Western numerals are the norm in commercial agriculture across the region.

WF-754 — Fonts covering Devanagari (Hindi), Bengali and Perso-Arabic (Pashto) are bundled with the application, not relied upon from the device.

WF-755 — Test on a physical Android 8 device in each script before sign-off.

WF-756 — Language switching takes effect immediately across the entire interface, without restarting the app.

WF-757 — Layouts accommodate a 40% expansion in string length without breaking. Emulator rendering is not evidence.

10.2 String catalogue

WF-758 — All strings live in an i18n platform with stable keys. No English text in application source. This includes push notification bodies, SMS text, WhatsApp template variables, report headings, PDF labels, error messages and unit abbreviations.

WF-759 — An **agronomic glossary** is maintained alongside the string catalogue, fixing the agreed translation of terms such as evapotranspiration, crop coefficient, pre-harvest interval, fertigation, canopy, vigour, water stress. These terms must be translated identically everywhere, and by someone who knows what they mean.

WF-760 — A named reviewer per language is recorded and signs off the catalogue before release.

WF-761 — Server-generated content — recommendation text, alert bodies, report sections — comes from the same catalogue as the app, keyed the same way.

10.3 Content in the user's language

WF-762 — Every notification, task instruction, report and recommendation is rendered in the **recipient's** language, resolved per user at send time.

WF-763 — Where a translation is missing for a key, fall back to English and log the missing key server-side. Never show a raw key to a user.

10.4 Locale conventions

WF-764 — Metric throughout. Temperature in Celsius. Water volume in cubic metres, with litres for per-tree figures and millimetres for depth.

WF-765 — Area is displayed in **dunum**, with hectares in brackets: `12.4 dunum (1.24 ha)`. The user may change the primary unit in Settings to hectares or acres, which swaps the order. 1 dunum = 0.1 ha throughout — the metric dunum.

WF-766 — Dates display in **both Gregorian and Hijri** when the app language is Arabic, Gregorian primary: `3 Aug 2026 (19 Safar 1448)`. The user may reverse the order or show one only. Hijri dates are display-only; all storage and calculation is Gregorian.

WF-767 — Prices display in the user's local currency — the GCC currencies and the Jordanian dinar — with the USD figure alongside: `SAR 375 / month (USD 100)`. Exchange rates come from the server and are refreshed daily. The app never calculates a conversion itself.

WF-768 — Time zones: all timestamps stored in UTC, displayed in the farm's local time zone, which is a property of the farm, not the device. A supervisor travelling must see farm-local times.

11 Offline behaviour

11.1 Principle

WF-780 — The app is **read-anywhere, write-in-the-field**. Everything the user has already seen remains viewable offline. A defined set of field actions can be created offline. Configuration and structural changes cannot.

11.2 What is cached

Data	Cached	Retention
Farm, block, field, plot records and geometry	✓	Until changed
Current crop cycles	✓	Until changed
Latest measure values per plot	✓	30 days
Latest measure raster for each plot, current measure only	✓	14 days
Map tiles covering the user's farms, zoom 12–17	✓	30 days
Tree records and latest health scores	✓	30 days
Tasks assigned to this user, plus all open tasks on their farms	✓	Until closed + 7 days
Active recommendations	✓	Until superseded
Weather: current + forecast at last sync	✓	3 days
The last 4 reports	✓	90 days
User guide	✓	Until updated
String catalogue for the user's language	✓	Until updated
Historical imagery beyond the latest	✗	—
Other users' tasks	✗	—

WF-781 — Total cache is capped at **500 MB** by default, user-adjustable between 100 MB and 2 GB in Settings. When the cap is reached, evict oldest imagery first, never queued user data.

WF-782 — Queued user data (photos, completions, observations) is **never evicted** and does not count against the cache cap.

11.3 What can be created offline

Action	Offline
Complete a task, with photo, quantity and note	✓
Create a field observation with photos	✓
Record an input applied (D7)	✓

Action	Offline
Read any cached screen	✓
Create or assign a task	✗
Edit a plot boundary	✗
Add or edit a crop cycle	✗
Invite a member or change a role	✗
Change subscription	✗

WF-783 — Actions that cannot be performed offline show a clear, calm message naming what is needed ("You need a connection to assign a task"), not a disabled control with no explanation.

11.4 Sync and conflict

WF-784 — Every offline-created record is written locally with a client-generated UUID and an idempotency key. Re-sending the same record must never create a duplicate.

WF-785 — Sync is automatic on regaining connectivity, on app foreground, and on a manual *Sync now*. It is incremental and resumable — a sync interrupted at photo 7 of 20 resumes at photo 7.

WF-786 — Conflict rules, per object type:

Object	Rule
Task completion	First write wins. If the task was already completed by someone else, the second completion is stored as an additional record and both are visible. Never silently discard a worker's evidence.
Field observation	Always accepted. Observations are additive; there is no conflict.
Input log	Always accepted, additive.
Task assignment or status changed server-side while offline	Server wins. The user is shown a non-blocking notice: "This task was changed while you were offline."

WF-787 — A sync failure is surfaced to the user in the connectivity indicator and in Settings → Data and storage, with the count of pending items and a retry. It is never silent.

WF-788 — Photos queued for upload survive the app being force-closed, the device being restarted and the app being updated. They are written to durable storage with their metadata before the user sees a confirmation.

WF-789 — Upload retries use exponential backoff, resume partial uploads, and prefer Wi-Fi where the user has set that preference — but a task completion older than 24 hours uploads on any connection regardless of the setting.

WF-790 — The offline store, including cached imagery and queued photos, is **encrypted at rest** on the device.

11.5 What the user sees

WF-791 — A connectivity indicator is present on every screen, with three visual states: online, offline, and syncing with a count of pending items.

WF-792 — Cached content older than 24 hours carries a visible age label.

12 Deployment and release

12.1 Repository and branching

WF-800 — Three long-lived branches: `develop` → `staging` → `main`. No commits directly to `staging` or `main`; everything arrives by pull request with at least one approving review.

WF-801 — Every pull request runs, and must pass: linting, unit tests, a build of both mobile platforms, and dependency scanning. A red build is not merged.

WF-802 — Semantic versioning `MAJOR.MINOR.PATCH`, with a monotonically increasing build number. The version and build number are displayed in the app (WF-315) and sent on every API request.

12.2 Configuration

WF-803 — No environment-specific value is compiled into the application except the API base URL, which is set per build flavour. Everything else — contact details, feature flags, minimum version, prices, support hours — is fetched from the backend at launch.

WF-804 — Three build flavours: `dev`, `staging`, `production`, installable side by side on the same device with distinct bundle identifiers, app names and icons.

WF-805 — Signing keys and store credentials live in the CI secret store, backed up in two separate locations. Losing the Android upload key makes the app permanently un-updatable.

12.3 Release checklist

Run this list for every release. Do not skip steps because the change was small.

Before the release branch is cut

1. All requirements in the release scope marked done and their acceptance criteria met.
2. String catalogue complete: zero missing keys in any of the five languages.
3. Named reviewer sign-off recorded for each language (WF-760).
4. RTL layout verified for Arabic and Pashto on a physical device.
5. Bengali, Hindi and Pashto rendering verified on a physical Android 8 device.
6. Offline scenarios tested: airplane mode capture, force-close during upload, sync after 24 hours offline, duplicate-submission test.
7. Entitlement matrix tested for all five plans, including locked-feature behaviour.
8. Role matrix tested for all three roles, including the server-side rejection of a client that tries to exceed its role.
9. Dependency scan clean of critical findings.
10. Crash-free session rate on staging above 99.5% over the last 7 days.

Cutting the release

11. Merge `develop` → `staging`, tag `vX.Y.Z-rc.N`.
12. Deploy backend to staging. Run database migrations. Confirm migrations are reversible.
13. Distribute the mobile build to TestFlight (internal) and Google Play internal testing.

14. UAT against the Jordanian farms (§13.2). Wafra sign-off required.

Going live

15. Merge `staging` → `main`, tag `vX.Y.Z`.

16. Deploy backend to production during an announced maintenance window, outside 05:00–19:00 Arabian Standard Time.

17. Run migrations. Verify with a smoke test of the 10 critical paths (§12.5) before releasing the mobile build.

18. Submit to App Store Review and Google Play. Expect 24–72 hours; see §12.4.

19. Staged rollout on Google Play: 10% → 50% → 100% over 72 hours, watching crash rate.

20. iOS phased release enabled.

21. Publish release notes in all five languages.

22. Tell Wafra support what changed, before users find out.

12.4 App store submission

WF-806 — Both store listings are prepared in **all five languages**: title, subtitle, description, keywords, screenshots and the privacy questionnaire.

WF-807 — Screenshots show the app in the language of that listing.

WF-808 — The privacy questionnaire (Apple Nutrition Label / Google Data Safety) is completed accurately for location, photographs, contact details, identifiers and usage data.

WF-809 — For review, provide a demo account populated with farms, imagery, advice and tasks. An empty app is rejected as non-functional.

WF-810 — Subscriptions are configured in App Store Connect and Play Console with: the free trial, localised prices per territory (SAR, AED, JOD, USD), and clear descriptions. Trial terms must appear in the app before purchase, as both stores require.

WF-811 — The app contains no link to an external purchase page for the subscriptions sold in-app (WF-332).

WF-812 — Location permission strings explain the actual purpose in the user's language, naming both uses: showing position on the farm map, and recording where field photographs were taken.

12.5 Smoke test — the ten paths

Run these against production after every deployment, on a real device, before announcing the release:

1. Sign up with a new mobile number, receive and enter the OTP.
2. Draw a farm boundary and save it; the farm appears on Home.
3. Open a plot; an measure map renders with a date and a legend.
4. Open Advice; a recommendation is listed with its reasoning.
5. Create a task from a recommendation and assign it to a second user.
6. As that second user, receive the notification in a different language and open the task.
7. In airplane mode, complete the task with a photograph. Confirmation appears immediately.
8. Restore connectivity; the completion and the photograph sync; the supervisor sees both.
9. Generate a weekly report; it downloads as a PDF in the requested language with Wafra branding.
10. Attempt a locked feature; the upgrade sheet appears and the API returns `402`.

WF-813 — A failure in any of the ten is a rollback, not a hotfix-in-place.

12.6 Rollback

WF-814 — Every backend deployment is rollback-capable within 15 minutes. Database migrations are written so that the previous application version continues to work against the new schema — additive first, destructive only in a later release.

WF-815 — Every material change is behind a server-side feature flag that can be turned off without an app update.

WF-816 — A forced-update mechanism driven by `minimum_supported_version` in `/config`, showing a blocking screen with a store link. Use it only for security fixes and breaking API changes.

13 Acceptance

13.1 Definition of done

WF-850 — A requirement is done when all of the following are true:

1. It behaves as specified on both platforms.
2. It behaves correctly in all five languages, including RTL.
3. It has an empty state, a loading state and an error state (WF-011, WF-012).
4. Its offline behaviour matches §11 — including "cannot be done offline", where that is the specification.
5. Its entitlement and capability behaviour is enforced server-side and verified by a test that calls the API directly, not through the app.
6. It works on a 360 × 640 dp screen at 200% font scale.
7. It has been used, once, by a person who did not build it.

13.2 User acceptance testing

WF-851 — UAT runs on staging against the **Jordanian farms already monitored by Wafra**, using real boundaries and real imagery history.

WF-852 — Two named sign-offs are required before production release:

Area	Sign-off
Agronomic correctness — advice, units, crop parameters, terminology in Arabic	Hani AlHabbab
Functional correctness — everything in this document	Romain Bigaré

14 Open items

Technical questions to close during the build. Each has a named owner.

#	Item	Owner
1	Data residency. Which jurisdiction client data must sit in, and what MMC can offer — see §A.1	Wafra
2	Named technical counterpart at MMC	MMC
3	WhatsApp Business API account and template approval started (WF-652)	Wafra
4	Intercrop analytics. How canopy/understory separation is produced, which crop pairings are supported, and what the app shows when separation is unavailable (WF-233, WF-235)	MMC
5	Purchase methodology. Whether in-app purchase is the right channel at all. Apple and Google take 15–30% of every subscription (15% under the small-business and second-year rates, 30% otherwise), both stores forbid linking to an external purchase page for the same product, and neither supports invoicing or annual contracts. The alternative is off-platform sale with the app reading a server-held entitlement flag — no store commission, no in-app purchase flow, but no self-serve signup either	Wafra
6	Domain and subdomain. Which Wafra-branded hostnames serve the API, the deep links and the web fallback pages, and who holds the DNS	MMC + Wafra
7	Source code and repository. Whether Wafra has read access to the repository, and what happens to the codebase, the build pipeline and the signing keys if the arrangement ends	MMC + Wafra
8	Acceptance sign-off (WF-852). Who signs off agronomic correctness and who signs off functional correctness. Proposal: Hani on agronomy, Romain on function. Should Mark be added as a third, commercial sign-off?	Wafra
9	Ownership of the registered-product register. This specification assumes MMC owns, sources and keeps current the plant protection product register for every country sold into (§§6.4). Wafra to settle internally whether it accepts that, or whether Wafra owns the register and supplies it as configuration — and who carries the consequence if an entry is wrong	Wafra
10	Territory-specific agronomic content. Who supplies crop coefficients, registered pesticide product lists, local thresholds and the terminology glossary, and who keeps them current thereafter (WF-283, WF-759)	Mark
11	Country-specific registered pesticide product lists for each market in the GCC and Jordan (WF-283)	Hani
12	Agronomic glossary and named translation reviewer per language (WF-759, WF-760)	Hani + Wafra
13	Brand assets: logo, colour tokens, app icon, splash, typeface with Latin, Arabic, Devanagari, Bengali and Perso-Arabic coverage, and store listing assets in five languages (WF-017)	Wafra
14	Terms of Use and Privacy Policy, reviewed against regional data protection law, in five languages	Wafra
15	Apple and Google developer accounts created and held by Wafra, with MMC granted access for deployment	Romain
16	Farm subdivision. Whether Farm → Block → Field → Plot is the right hierarchy, whether both intermediate levels are needed, and whether the platform can support it at all — it is one crop per plot today (§§2.2, §§5.4.4)	Wafra + MMC
17	Combined plan pricing. A client holding crops and trees buys one Complete plan (§§9.1.1), but crop capability is priced per farm or hectare and tree capability per tree. How the combined price is derived, and what the supplier charges Wafra for it	Wafra

#	Item	Owner
18	Tree Pro scope. This specification puts personalised AI advisory, yield estimation, orchard yield optimisation, plot-level harvest planning, scouting workflows, hyper-local weather, custom weather alerts, higher-resolution satellites, canopy structure and the detailed monthly report into Tree Pro (\$\$9.4). All ten sit in the supplier's Advanced tree tier today. Agree the commercial change, or cut them from the specification	Wafra
19	Fertigation scheduler and NPK reports. Both sit above the tier Crop Advanced maps to (\$\$9.3). Bring them in, or accept irrigation-only scheduling	Wafra

15 Assumptions

Stated so they can be corrected rather than discovered. If any is wrong because the capability already exists, say so in the feasibility response and the estimate should fall accordingly.

1. The existing crop and tree analytics are available through an internal interface the app can consume, at the tier levels described in §9. This document does not ask for new models.
2. Task management, the advised-vs-applied record and the report at WF-319 are net-new development.
3. The farm → block → field → plot hierarchy with multiple crops per farm is a change to the current data model, which supports one crop per plot.
4. Farm sharing and role-based access is a change to the current product, which is single-user.
5. Five-language support with full RTL is substantially new; the current product is effectively English-only.
6. Offline capture as specified in §11 is new.
7. Entitlement, trial and in-app purchase infrastructure is new for this branded product.
8. Nothing in this document requires changes to satellite acquisition or to the AI models.

Annex A — Hosting, data and security

A.1 Data residency

TBD — Wafra.

A.2 Backups

WF-900 — Daily full backup, point-in-time recovery to any moment in the last 7 days, and a restore tested at least quarterly with the result recorded.

WF-901 — Backups and disaster-recovery replicas are held under the same residency terms as primary storage.

WF-902 — The system supports export of all data belonging to a Wafra client, in a documented, machine-readable format, on request, within 10 working days.

A.3 Media and personal data

WF-903 — Photographs are served through short-lived signed URLs, never public object storage links. A leaked URL must stop working.

WF-904 — Every task photograph is tagged with a named worker, a timestamp and a GPS coordinate, which constitutes location processing of identified individuals. Required consequences:

- The purpose is stated in the privacy policy in all five languages.
- Workers see, at first launch, a plain-language notice explaining that photographs they take record their location and are visible to their supervisor and the farm owner.
- Location is recorded **only** at the moment a photograph or observation is captured. The app does not track worker position continuously.
- A worker can request deletion of their personal data.

WF-905 — Personal data is not used for any purpose beyond operating the service without a separate consent.

WF-906 — Log records containing personal data are excluded from application logs and error-reporting tools. Mobile numbers, names and coordinates must not appear in crash reports.

Annex B — Requirement register

356 requirements. Section numbers refer to this document.

ID	§	Requirement
WF-001	2.3	Mobile application for iOS 15+ and Android 8.0 (API 26)+. No browser application in this release
WF-002	2.3	Usable on a 5-inch screen at 360 × 640 dp
WF-003	2.3	Install size below 80 MB
WF-004	2.3	Minimum touch target 48 × 48 dp, with at least 8 dp between adjacent targets
WF-005	2.3	The primary action on every screen sits in the bottom third of the display
WF-006	2.3	Body text no smaller than 16 sp
WF-007	2.3	Respect the operating-system font-size setting up to 200% without clipping text or losing...
WF-008	2.3	Status is never communicated by colour alone
WF-009	2.3	Health and status use this four-state scale everywhere, with the same icon, word and colour...
WF-010	2.3	No screen presents more than one primary action
WF-011	2.3	Every screen that can legitimately have nothing to show — no farms yet, no plots in a block,...
WF-012	2.3	Every screen has an explicit loading state and an explicit error state with a plain-language...
WF-013	2.3	Numbers are always shown with their unit, rounded to actionable precision: "412 m³" not...
WF-014	2.3	Icons carry a text label beneath or beside them
WF-015	2.3	Never block the user with a full-screen "no connection" message when cached data exists
WF-016	2.3	Every data-entry action that can be performed offline shows immediate local confirmation
WF-017	2.3	The application carries Wafra branding only — logo, colours, app icon, splash, store listing...
WF-030	3.1	Owner and Supervisor see five tabs:
WF-031	3.1	Worker sees three tabs, and lands on My Work at launch:
WF-032	3.1	A badge on the Advice tab shows the count of unread recommendations at Action needed or...
WF-033	3.1	A badge on the Tasks / My Work tab shows the count of tasks assigned to this user and due...
WF-034	3.1	Tab bar labels are always visible
WF-100	4.2	Registration creates an account with no role
WF-101	4.2	A user who creates a farm becomes a Farm Owner of it
WF-102	4.2	A user who accepts an invitation receives the role carried in that invitation — Farm...
WF-103	4.2	The same person may do both
WF-104	4.2	A Worker can never create, delete or edit a farm, a plot or a boundary, on any farm, however...
WF-105	4.2	An invitation is single use, expires after 7 days, and is bound to the role and the farm...
WF-106	4.2	Four delivery methods, all producing the same single-use token: WhatsApp deep link...
WF-107	4.2	An Owner sees all pending invitations for their farms with expiry and a one-tap revoke
WF-108	4.2	Where a Worker has no smartphone of their own, a Supervisor can complete the invitation on...
WF-109	4.3	Each language is shown in its own script, with the English name beside it
WF-110	4.3	Pre-selected to the device's OS language if it is one of the five; otherwise English
WF-111	4.3	Selecting a right-to-left language mirrors the layout immediately, on this screen, before...
WF-112	4.3	Never shown again after first run
WF-113	4.4	Three cards maximum, skippable from any card
WF-114	4.5	Registration is by mobile number with SMS one-time code
WF-115	4.5	Country selector defaults from the SIM's country, then device locale, then Saudi Arabia
WF-116	4.5	The number field accepts spaces and dashes and normalises them

ID	§	Requirement
WF-117	4.5	Terms and Privacy links open in-app, in the user's selected language
WF-118	4.5	Six-digit code, valid for 10 minutes
WF-119	4.5	Resend available after 45 seconds
WF-120	4.5	After 5 incorrect code entries the account is locked for 15 minutes and the user is shown...
WF-121	4.6	Name is mandatory; email optional
WF-122	4.6	A show/hide control on the password field
WF-123	4.6	Password is optional for Worker accounts created by invitation; those may sign in by SMS...
WF-124	4.7	This screen routes, it does not assign privileges. "I manage my own farm" continues to A7...
WF-125	4.7	A user who arrives through an invitation deep link skips A6 entirely
WF-126	4.7	I just want to tour the app opens a demo session: a read-only example farm with sample...
WF-127	4.7	A demo session writes nothing to the server, creates no account, consumes no entitlement and...
WF-128	4.7	Every action in a demo session that would normally write data completes locally and visibly...
WF-129	4.7	The demo fixture is delivered by the backend configuration endpoint so its content can be...
WF-130	4.8	Three options — Field crops, Trees and orchards, Both — setting the default farm_type for...
WF-131	4.9	This screen is reachable only by Owners and Supervisors with the plot.create capability
WF-132	4.9.1	Opens centred on the device's GPS position at approximately 1:5,000
WF-133	4.9.1	Satellite basemap by default
WF-134	4.9.1	Live area readout in dunum with hectares in brackets
WF-135	4.9.1	Undo removes the last placed vertex
WF-136	4.9.1	Vertices are draggable with a touch target of at least 48 dp, offset above the fingertip so...
WF-137	4.9.1	Self-intersecting polygons are rejected with a plain-language message and the offending...
WF-138	4.9.1	A polygon smaller than 0.1 ha or larger than 10,000 ha triggers a confirmation, not a rejection
WF-139	4.10	Only farm name and farm type are mandatory
WF-140	4.10	Irrigation system options: Drip, Centre pivot, Sprinkler, Bubbler, Flood/furrow, Other, Not...
WF-141	4.10	Soil type may be left blank; the backend fills it from the soil layer where available and...
WF-142	4.11	The plan chooser shows only the plan family matching the farm type just created
WF-143	4.11	Plan order is always Basic → Pro → Advanced for crops and Basic → Pro for trees
WF-144	4.11	Prices are fetched from the server, never hard-coded, and displayed in the currency of the...
WF-145	4.11	Compare all features opens the full comparison table (F6)
WF-146	4.11.1	Where the account has at least one crop farm and at least one tree farm, the plan chooser...
WF-147	4.11.1	A combined plan is a single purchasable product with a single price and a single renewal date
WF-148	4.11.1	A mixed farm — crops and trees in different plots of the same farm — requires a combined plan
WF-149	4.11.1	Where a client on a crop-only plan adds their first tree farm, the app offers an upgrade to...
WF-150	4.11.1	Entitlement is resolved server-side as a single set of feature keys
WF-151	4.11.1	Downgrading from a combined plan to a single family is only possible once the farms of the...
WF-152	4.12	A confirmation screen stating when the first satellite imagery will be available, and a...
WF-153	4.13	Opening an invitation deep link, with or without the app installed, leads to: install if...
WF-154	4.13	An invited user never sees A6, A7–A11 or A12
WF-155	4.13	Invitations expire after 7 days and are single use
WF-156	4.13	A used, expired or revoked invitation shows a clear message and a Contact the farm owner action
WF-157	4.13	On successful join, a Worker lands directly on My Work; a Supervisor lands on Home filtered...
WF-158	4.14	Login accepts mobile number + password, or mobile number + SMS code
WF-159	4.14	A language control is present on the login screen
WF-160	4.14	Sessions persist for 90 days of activity on a personal device
WF-161	4.14	A shared device mode toggle in Settings shortens the session to 12 hours and requires...

ID	§	Requirement
WF-162	4.14	Log out is always available and clears the local cache including cached imagery and queued...
WF-163	4.15	A demo session opens a fixture: two example farms — one crop, one tree — with plots, imagery...
WF-164	4.15	A demo session writes nothing to the server, creates no account, consumes no entitlement,...
WF-165	4.15	The banner at the top of every screen reads Demo — nothing here is saved and carries an Exit...
WF-166	4.15	Every action that would normally write data completes locally and visibly, then resets with...
WF-167	4.15	Any action that genuinely requires an account — subscribing, inviting a team member, adding...
WF-168	4.15	The session is bound to the app process
WF-169	4.15	All plan-gated features are shown unlocked in the demo, so a prospect sees what the product...
WF-170	4.15	The demo fixture is delivered by the backend configuration endpoint, so its content can be...
WF-200	5.1	The summary card at the top counts farms by worst-plot status, not by average
WF-201	5.1	See what to do navigates to the Advice inbox (D1) pre-filtered to Action needed and Urgent
WF-202	5.1	Each farm card shows: status icon and colour, farm name, farm type icon, area, plot or tree...
WF-203	5.1	"Updated X ago" refers to the age of the most recent satellite imagery, not the last app refresh
WF-204	5.1	Farms are sorted by severity first, then by most recently viewed
WF-205	5.1	Pull to refresh, with a visible refresh button as the non-gesture equivalent
WF-206	5.1	Empty state: "You have no farms yet" with an Add your first farm button leading to A7
WF-207	5.2	The health block lists only measures included in the farm's current plan
WF-208	5.2	Tapping any row in the health block opens the measure viewer (B7) for that measure across...
WF-209	5.2	The Trees row is present only for tree and mixed farms
WF-210	5.2	The weather strip shows current conditions and a 7-day or 14-day forecast according to plan...
WF-211	5.2	Farm settings (⚙, top right) is visible to Owner and Supervisor only
WF-212	5.3	Plots are grouped by block where blocks exist; where a farm has no blocks, the grouping...
WF-213	5.3	Sort options: Needs attention first (default), Name, Area, Most recently changed
WF-214	5.3	Each plot row shows the measure value and its change over the last 7 days, with a direction...
WF-215	5.4	The measure selector is a dropdown listing every measure available on this farm's plan, in...
WF-216	5.4	Date stepping with ◀▶ moves between available imagery dates, not calendar days
WF-217	5.4	The interpretation line below the map names where and how long, not only what: "the east..."
WF-218	5.4	See what to do is the primary action and links to the relevant recommendation (D2–D5)
WF-219	5.4	The ⋮ menu contains: Edit plot, Edit boundary, Add crop cycle, Add observation, Create task,...
WF-220	5.4.1	Full-screen measure map, pinch-zoom and pan, with a persistent legend showing the value...
WF-221	5.4.1	Tapping any point on the map shows the value at that point in a small callout, with its...
WF-222	5.4.1	The legend scale is fixed per measure, not auto-scaled to the current image, so that...
WF-223	5.4.2	Side-by-side or swipe-divider comparison of the same plot on two dates, with both dates...
WF-224	5.4.2	A compare with previous years mode showing the same calendar week across up to 5 previous years
WF-225	5.4.3	Adding a new crop cycle to a plot that already has an open one prompts the user to close the...
WF-226	5.4.3	Closing a crop cycle never deletes it
WF-227	5.4.3	Crop cycle fields: crop, variety, sowing/planting date, expected harvest date, actual...
WF-228	5.4.3	The crop picker is searchable, grouped by category (cereals, forage, vegetables, fruit...
WF-229	5.4.4	A farm supports many plots with different crops, varieties and planting dates simultaneously
WF-230	5.4.4	The hierarchy Farm → Block → Field → Plot is supported in the data model
WF-231	5.4.4	A user may create, rename, reorder and delete blocks and fields
WF-232	5.4.5	A plot may record a primary crop and one or more secondary crops (for example date palm with...
WF-233	5.4.5	Measure values and health status on an intercropped plot are attributed per crop, using the...
WF-234	5.4.5	Recommendations are generated per crop within the plot
WF-235	5.4.5	Where separation cannot be computed for a given date — insufficient resolution, cloud, or an...

ID	§	Requirement
WF-236	5.4.6	Editable: farm name, farm type, address/region, irrigation system, soil type, default...
WF-237	5.4.6	Contains Transfer ownership (Owner only, with SMS confirmation) and Delete farm (Owner only,...
WF-238	5.5	The screen leads with the status distribution; the list below is filtered to what needs...
WF-239	5.5	Filters: status, plot, row, species, planting year, and "declining" (health falling over the...
WF-240	5.5	A bulk action from any filtered list: create one task covering all matching trees
WF-241	5.5	Tree detail (B10) shows: tree ID, QR code, species and variety, planting year and age,...
WF-242	5.5	QR code scanning from the Map tab or the tree list opens that tree's detail screen directly
WF-243	5.5	Missing and dead trees are shown as a distinct state with their own count, not folded into...
WF-244	5.5	Where a tree-level measure is not in the client's plan, it is listed and locked rather than...
WF-245	5.6	Estimated yield is shown for the orchard and per tree, with a confidence range and the...
WF-246	5.6	Ripeness distribution is shown as a proportion of the trees in each stage, for the whole...
WF-247	5.6	The pick order lists plots ranked by readiness, each with its ripe proportion, an estimated...
WF-248	5.6	Create harvest tasks generates one task per plot in the pick order, pre-filled with the...
WF-249	5.6	Yield optimisation advice appears as a single plain-language line above the action, in the...
WF-250	5.6	Where the plan does not include harvest planning, the entry point is visible and locked (§9.5)
WF-251	5.7	On Tree Pro, the advice inbox additionally carries orchard-level recommendations generated...
WF-252	5.7	A personalised recommendation states what it learned from, in one line: "based on three...
WF-253	5.8	The map opens on the user's farms, zoomed to fit all of them, or to the currently selected...
WF-254	5.8	Plot polygons are filled with the currently selected measure and carry their status icon and...
WF-255	5.8	Tapping a plot opens a bottom sheet (C3) with: plot name, crop, area, current measure value...
WF-256	5.8	Layer picker (C2) offers, subject to plan:
WF-257	5.8	Layer selections persist between sessions
WF-258	5.8	Locked layers appear in the list with a lock icon and open the upgrade screen when tapped
WF-259	5.8	The user's own GPS position is shown when location permission is granted, with a Locate me...
WF-260	5.8	Date stepper on the map moves through available imagery dates for the visible plots, as WF-216
WF-261	5.8	Compare mode (C4) puts a draggable vertical divider across the map with a different date...
WF-262	5.8	Map tiles for the user's own farms are cached for offline use (§11.2)
WF-263	5.8	Search accepts farm name, plot name, crop name and tree ID
WF-264	5.8.1	Editing an existing boundary uses the same interaction as A8: tap to add, drag to move, and...
WF-265	5.8.1	A boundary change is a versioned event, not an overwrite
WF-266	5.8.1	Boundary editing requires the farm.boundary.edit capability (Owner and Supervisor)
WF-267	5.9	Recommendations are grouped by urgency into Today / This week / Later, and within each group...
WF-268	5.9	Every recommendation card states, in this order and in plain language: what to do, how much,...
WF-269	5.9	Every recommendation carries two actions inline, without opening the detail screen: Create...
WF-270	5.9	Filters: farm, plot, type (irrigation, nutrition, protection, weather, harvest), status...
WF-271	5.9	A recommendation is marked done when the user records an action against it (D7) or completes...
WF-272	5.9	Recommendations are never silently deleted or replaced
WF-273	5.9	Where the client's plan does not include advisory, the Advice tab is still present and shows...
WF-274	5.9.1	The headline number is shown in three units at once: depth (mm), total volume (m³), and —...
WF-275	5.9.1	Pumping time is calculated and displayed where the plot has a recorded system flow rate
WF-276	5.9.1	The "Why" section lists every input used, with its value
WF-277	5.9.1	Assumptions we used is editable
WF-278	5.9.1	The disclaimer line is present on every advisory detail screen, in the user's language, and...
WF-279	5.9.2	Same structure: headline dose, split across application windows, the reasoning, editable...
WF-280	5.9.2	Nutrient recommendations are expressed as elemental N, P, K, Ca, Mg per hectare, with an...

ID	§	Requirement
WF-281	5.9.2	Where MMC's platform provides fertiliser insight but not a full fertigation schedule for the...
WF-282	5.9.3	Recommendations lead with the active ingredient and rate, not a commercial product name
WF-283	5.9.3	Product names are drawn from the registered-product register for the user's country — see §6.4
WF-284	5.9.3	Every crop protection recommendation displays the pre-harvest interval for the user's...
WF-285	5.9.3	A permanent, non-dismissible notice: Check the product label and your local regulations...
WF-286	5.9.3	The screen includes symptom photographs and a short identification guide from MMC's disease...
WF-287	5.9.3	Every crop protection recommendation is written to the advisory log when it is generated —...
WF-288	5.9.4	Alert types: frost, heat stress, high wind (spraying), heavy rain, sandstorm/dust, high...
WF-289	5.9.4	Weather alerts are delivered as push notifications at the criticality defined in §7.2, in...
WF-290	5.9.5	Recording an action takes at most three taps from the advice card for the common case
WF-291	5.9.5	The recorded action writes to the plot's activity history and the farm diary, and is...
WF-292	5.9.5	Once actions have been recorded for a plot, the plot detail screen shows an...
WF-293	5.9.5	This screen works offline
WF-294	5.10	Task states: Open → In progress → Done, plus Cancelled
WF-295	5.10	A task has: title, description, type, farm, plot or tree selection, assignee, due date and...
WF-296	5.10	Task types (used for icons, filtering and reporting): irrigation, fertiliser, spraying,...
WF-297	5.10	A task may be assigned to one person
WF-298	5.10	A task may cover many plots or many trees — see WF-240
WF-299	5.10	Overdue tasks are always shown first, always visually distinct
WF-300	5.10.1	Creating a task from a recommendation pre-fills title, description (including what and how...
WF-301	5.10.1	The assignee picker lists only users who have access to the selected farm, shows each...
WF-302	5.10.1	On save, the assignee is notified through their configured channel (§7) in their language,...
WF-303	5.10.2	The Worker view of a task is stripped to: what, how much, where, when, and one big button
WF-304	5.10.2	Show me where opens the map centred on the target plot or tree, with a line and a distance...
WF-305	5.10.2	Mark as done opens E4:
WF-306	5.10.2	Everything on the completion screen is optional except the confirm tap
WF-307	5.10.2	Photographs are automatically stamped with GPS coordinates, timestamp, task ID, plot ID and...
WF-308	5.10.2	The whole completion flow works offline, confirms locally and immediately, and syncs later (§11)
WF-309	5.10.2	I cannot do this records a reason from a short picture-supported list and notifies the task...
WF-310	5.10.3	A standalone observation can be captured from the Tasks tab, the map, a plot or a tree,...
WF-311	5.10.3	Observations appear on the map as pins, on the plot's activity history, and in the farm diary
WF-312	5.10.3	Where the plan includes photo-based disease detection (E7), photographing a leaf returns an...
WF-313	5.10.3	Works fully offline
WF-314	5.11	Menu contents are filtered by role
WF-315	5.11	The version and build number are always visible on this screen
WF-316	5.11.1	Reports are generated server-side as PDF, never assembled on the device
WF-317	5.11.1	Every report is produced in the language the user requests, from the same string catalogue...
WF-318	5.11.1	Reports are shareable through the OS share sheet (WhatsApp, email, file save)
WF-319	5.11.1	The Irrigation: advised vs applied report is not in the existing report set and must be built
WF-320	5.11.1	Reports outside the client's plan appear locked, not hidden
WF-321	5.11.1	The weekly report is also delivered by email to any user with an email address and by...
WF-322	5.11.2	Pending invitations are listed with their expiry and a one-tap cancel
WF-323	5.11.2	Each member row shows their role and their app language, so the assigner knows which...
WF-324	5.11.2	Invite (F3) is by mobile number (primary), email, link, 6-character code or QR — see §4.1
WF-325	5.11.2	Invitations may be scoped to specific farms, not necessarily the whole account

ID	§	Requirement
WF-326	5.11.2	QR-code invitations produce single-use, short-expiry, role-scoped tokens
WF-327	5.11.2	Removing a member takes effect immediately, invalidates their active sessions for that farm,...
WF-328	5.11.3	Crop and tree subscriptions are separate and displayed separately
WF-329	5.11.3	Trial status shows days remaining, prominently, from day one
WF-330	5.11.3	Purchase and renewal are handled through Apple In-App Purchase and Google Play Billing
WF-331	5.11.3	Entitlement is always determined by the server, never by the local receipt alone
WF-332	5.11.3	The Contact Wafra route for invoicing and annual contracts is informational only — it opens...
WF-333	5.11.4	Language and region (F8): app language, area unit preference (dunum/hectare/acre — dunum...
WF-334	5.11.4	Notifications (F9): per-channel and per-category toggles, quiet hours, and a channel...
WF-335	5.11.4	Data and storage (F10): current cache size, "download imagery on Wi-Fi only" toggle, "clear...
WF-336	5.11.4	A shared device toggle (WF-161)
WF-337	5.11.4	Settings includes Delete my account, which explains what will be deleted and what will be...
WF-338	5.11.5	Owner-only
WF-339	5.11.5	The activity log is append-only
WF-340	5.11.6	Help contains a searchable user guide, in all five languages, with short illustrated articles
WF-341	5.11.6	Contact Wafra presents exactly two large buttons plus a support ticket option:
WF-342	5.11.6	WhatsApp opens a chat with +966 54 810 0443, pre-filled with a short message containing the...
WF-343	5.11.6	Email opens the device mail client to info@wafragreen.com, with the same diagnostic details...
WF-344	5.11.6	Both destinations are configuration values fetched from the backend, not compiled into the...
WF-345	5.11.6	The support ticket route uses the existing ticketing capability where the plan includes it
WF-600	6.3	Every recommendation is written to an advisory log at the moment it is generated, before it...
WF-601	6.3	The advisory log is write-once
WF-602	6.3	Retention is 7 years
WF-603	6.3	A superseded recommendation is marked superseded and linked to the one that replaced it
WF-604	6.3	The log is exportable, on request, for a named client and date range, in a documented...
WF-605	6.3	Acknowledgement and action are recorded against the log entry: who saw the advice, when, and...
WF-606	6.4	A register of plant protection products, held per country, containing at minimum: active...
WF-607	6.4	A product is only shown to a user whose farm is in a country where that product is registered
WF-608	6.4	Recommendations lead with the active ingredient and rate
WF-609	6.4	The pre-harvest interval is taken from the register for the user's country, and the earliest...
WF-610	6.4	Register entries are versioned with an effective-from date
WF-611	6.4	The register is updated server-side and takes effect without an app release
WF-612	6.4	Every change to the register is recorded with who changed it, when, and what changed
WF-613	6.4	Where an entry has not been verified within 12 months, product names from that entry are...
WF-614	6.5	Every recommendation carries a rule_version
WF-615	6.5	Every input used is stored with the recommendation, with its value, unit and source
WF-616	6.5	Assumptions a user can correct — irrigation efficiency, soil type, system flow rate — are...
WF-617	6.6	Where the client's plan does not include a given advisory capability, no recommendation of...
WF-618	6.6	Where canopy separation cannot be computed for an intercropped plot on a given date,...
WF-619	6.6	Where a mandatory input is missing, no recommendation is generated with a substituted value
WF-620	6.6	Crop protection recommendations above a defined risk threshold — rate at the top of the...
WF-621	6.7	Every advisory detail screen carries a non-dismissible line stating that the recommendation...
WF-622	6.7	Every crop protection recommendation carries a non-dismissible instruction to check the...
WF-623	6.7	The Terms of Use state that recommendations are advisory, that the farmer remains...
WF-624	6.7	Advice is never presented as certain where it is not

ID	§	Requirement
WF-650	7.1	Three channels: push notification, WhatsApp, and SMS
WF-651	7.1	Every notification is sent in the recipient's saved language, not the sender's, not the...
WF-652	7.1	WhatsApp delivery uses the WhatsApp Business API with pre-approved message templates
WF-653	7.1	SMS is the fallback where WhatsApp delivery fails or the recipient has not opted in
WF-654	7.2	Categories marked "no" above are safety- or contract-critical and may not be switched off...
WF-655	7.2	Quiet hours are user-configurable, default 21:00–05:00 local, and apply to every category...
WF-656	7.2	Every notification carries a deep link that opens the exact object it concerns — task,...
WF-657	7.2	Notification volume is capped at 6 per user per day, excluding task assignments and severe...
WF-670	8.1	Permissions are implemented as a capability matrix
WF-671	8.1	Capabilities are the unit of enforcement, not role names
WF-672	8.2	A Supervisor may invite Workers but not other Supervisors or Owners
WF-673	8.2	Access is granted per farm, not per account
WF-674	8.2	A farm may have more than one Owner
WF-675	8.3	Every capability is enforced server-side
WF-676	8.3	Session tokens carry the user identity only
WF-677	8.4	Invitation tokens are single use, expire after 7 days, are bound to the role and farm scope...
WF-678	8.4	An invitation link that has been used, expired or revoked shows a clear message and a...
WF-700	9.1	Two plan families, and a combined family for clients holding both:
WF-701	9.1	Plan names in the interface are exactly Basic, Pro, Advanced
WF-702	9.1	The mapping from a plan to the underlying platform tier is a server-side configuration, not...
WF-703	9.1.1	An account holds one plan at a time, whatever mix of farms it contains
WF-704	9.1.1	The plan family offered is derived from the account's farms, not chosen by the user: crop...
WF-705	9.1.1	Adding a farm of the other type to a single-family account puts that farm in a pending...
WF-706	9.1.1	Archiving every farm of one type offers a downgrade from Complete to the remaining single...
WF-707	9.1.1	Trees are counted for entitlement at the account level across all tree farms, and crop area...
WF-708	9.2	Every new account receives one free month on the plan they select at A12
WF-709	9.2	Trial state is held server-side
WF-710	9.2	Trial expiry warnings at 7 days, 3 days and 1 day remaining, by push and email
WF-711	9.2	On trial expiry without a subscription, the account enters read-only state: the user can...
WF-712	9.5	A feature outside the user's plan is visible and locked, never hidden, unless it would be...
WF-713	9.5	A locked feature shows a lock icon in place of its value or content
WF-714	9.5	The upgrade sheet names the plan required and describes the benefit in one sentence of plain...
WF-715	9.5	Entitlement checks happen server-side on every request
WF-750	10.1	Five languages ship at launch: English, Arabic, Hindi, Bengali, Pashto
WF-751	10.1	Arabic and Pashto are both right-to-left
WF-752	10.1	RTL treatment includes: layout mirroring, navigation and back-arrow direction, list...
WF-753	10.1	Numbers use Western Arabic numerals (0–9) in all languages including Arabic, unless the user...
WF-754	10.1	Fonts covering Devanagari (Hindi), Bengali and Perso-Arabic (Pashto) are bundled with the...
WF-755	10.1	Test on a physical Android 8 device in each script before sign-off
WF-756	10.1	Language switching takes effect immediately across the entire interface, without restarting...
WF-757	10.1	Layouts accommodate a 40% expansion in string length without breaking
WF-758	10.2	All strings live in an i18n platform with stable keys
WF-759	10.2	An agronomic glossary is maintained alongside the string catalogue, fixing the agreed...
WF-760	10.2	A named reviewer per language is recorded and signs off the catalogue before release
WF-761	10.2	Server-generated content — recommendation text, alert bodies, report sections — comes from...

ID	§	Requirement
WF-762	10.3	Every notification, task instruction, report and recommendation is rendered in the...
WF-763	10.3	Where a translation is missing for a key, fall back to English and log the missing key...
WF-764	10.4	Metric throughout
WF-765	10.4	Area is displayed in dunum, with hectares in brackets: 12.4 dunum (1.24 ha)
WF-766	10.4	Dates display in both Gregorian and Hijri when the app language is Arabic, Gregorian...
WF-767	10.4	Prices display in the user's local currency — the GCC currencies and the Jordanian dinar —...
WF-768	10.4	Time zones: all timestamps stored in UTC, displayed in the farm's local time zone, which is...
WF-780	11.1	The app is read-anywhere, write-in-the-field
WF-781	11.2	Total cache is capped at 500 MB by default, user-adjustable between 100 MB and 2 GB in Settings
WF-782	11.2	Queued user data (photos, completions, observations) is never evicted and does not count...
WF-783	11.3	Actions that cannot be performed offline show a clear, calm message naming what is needed...
WF-784	11.4	Every offline-created record is written locally with a client-generated UUID and an...
WF-785	11.4	Sync is automatic on regaining connectivity, on app foreground, and on a manual Sync now
WF-786	11.4	Conflict rules, per object type:
WF-787	11.4	A sync failure is surfaced to the user in the connectivity indicator and in Settings → Data...
WF-788	11.4	Photos queued for upload survive the app being force-closed, the device being restarted and...
WF-789	11.4	Upload retries use exponential backoff, resume partial uploads, and prefer Wi-Fi where the...
WF-790	11.4	The offline store, including cached imagery and queued photos, is encrypted at rest on the...
WF-791	11.5	A connectivity indicator is present on every screen, with three visual states: online,...
WF-792	11.5	Cached content older than 24 hours carries a visible age label
WF-800	12.1	Three long-lived branches: develop → staging → main
WF-801	12.1	Every pull request runs, and must pass: linting, unit tests, a build of both mobile...
WF-802	12.1	Semantic versioning MAJOR.MINOR.PATCH, with a monotonically increasing build number
WF-803	12.2	No environment-specific value is compiled into the application except the API base URL,...
WF-804	12.2	Three build flavours: dev, staging, production, installable side by side on the same device...
WF-805	12.2	Signing keys and store credentials live in the CI secret store, backed up in two separate...
WF-806	12.4	Both store listings are prepared in all five languages: title, subtitle, description,...
WF-807	12.4	Screenshots show the app in the language of that listing
WF-808	12.4	The privacy questionnaire (Apple Nutrition Label / Google Data Safety) is completed...
WF-809	12.4	For review, provide a demo account populated with farms, imagery, advice and tasks
WF-810	12.4	Subscriptions are configured in App Store Connect and Play Console with: the free trial,...
WF-811	12.4	The app contains no link to an external purchase page for the subscriptions sold in-app (WF-332)
WF-812	12.4	Location permission strings explain the actual purpose in the user's language, naming both...
WF-813	12.5	A failure in any of the ten is a rollback, not a hotfix-in-place
WF-814	12.6	Every backend deployment is rollback-capable within 15 minutes
WF-815	12.6	Every material change is behind a server-side feature flag that can be turned off without an...
WF-816	12.6	A forced-update mechanism driven by minimum_supported_version in /config, showing a blocking...
WF-850	13.1	A requirement is done when all of the following are true:
WF-851	13.2	UAT runs on staging against the Jordanian farms already monitored by Wafra, using real...
WF-852	13.2	Two named sign-offs are required before production release:
WF-900	A.2	Daily full backup, point-in-time recovery to any moment in the last 7 days, and a restore...
WF-901	A.2	Backups and disaster-recovery replicas are held under the same residency terms as primary...
WF-902	A.2	The system supports export of all data belonging to a Wafra client, in a documented,...
WF-903	A.3	Photographs are served through short-lived signed URLs, never public object storage links
WF-904	A.3	Every task photograph is tagged with a named worker, a timestamp and a GPS coordinate, which...

ID	§	Requirement
WF-905	A.3	Personal data is not used for any purpose beyond operating the service without a separate...
WF-906	A.3	Log records containing personal data are excluded from application logs and error-reporting...